

Complementation and Common Ground: Discursive effects in Biblical Hebrew

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Abstract

Clausal connectives, such as complementizers or circumstantial markers, are highly grammaticalized functional elements. This article addresses the question whether part of their original meaning persists after grammaticalization has occurred: can a seemingly neutral clausal connective be semantically or pragmatically charged due to persisting meaning? It does this by discussing a case study of a highly polysemous clausal connective in Biblical Hebrew, כִּי *kī*. In particular, I argue that this clausal connective presupposes that the information content it introduces is in the Common Ground. Thus, *kī* is used in three contexts: (a) to refer to discourse-old information, (b) to introduce information that the Addressee can accommodate, and (c) to present discourse-new information as part of the Common Ground, for discursive effect. The many different uses of *kī* (introducing object and subject clauses as well as causal, temporal, conditional, adversative, concessive, and resultative circumstantial clauses) can then be derived from contextual clues given the general function of marking Common Ground. Furthermore, I argue that this function of *kī* is related to its origin as a [+distal] deictic lexeme, for which I draw a parallel with English *that*. The analysis thus adds to a growing body of evidence for the possibility of employing referential features in the left periphery to express relations between interlocutors on the one hand and information content on the other.

Keywords: complementizers; information status; Common Ground; [+distal]; Biblical Hebrew

1 Introduction

Cross-linguistically, clausal connectives can develop from a multitude of sources.¹ Kuteva et al. (2019) list instances of complementizers derived from allative particles, demonstrative pronouns, question words, quotative markers, relative pronouns, similitive particles, light nouns ('matter', 'thing'), and verbs meaning 'resemble' or 'say'. In some cases, these developments go through long grammaticalization chains, such as ALLATIVE > PURPOSE > INFINITIVE > COMPLEMENTIZER (Kuteva et al. 2019: 49). A relevant question is then to what degree the original meaning of these highly grammaticalized elements has bleached away. If some interpretative traces remain, the choice for one or another clausal connective in a certain context may have a semantic or pragmatic impact.

In this article I argue, on the basis of an in depth corpus study of Biblical Hebrew, that clausal connectives originating from a deictic particle can indeed carry such interpretative traces. To clarify the proposal, I first briefly introduce the idea with English *that* (for more details, as well as discussion of Romance complementizers, see section 1.2 below and Staps & Rooryck 2023). Consider the following examples:

- (1) a. *Bio industry is still allowed.*
b. *That bio industry is still allowed!* (Staps & Rooryck 2023: 1204)
- (2) a. *I always believed (that) the jury was bribed.*
b. **(That) the jury was bribed, I always believed.* (Staps & Rooryck 2023: 1209)

Note that while (1a) can be used to convince the Addressee of the fact that bio industry is still allowed, (1b) cannot. A Speaker uttering (1b) assumes that its propositional content is in the Common Ground—that is, that Speaker and Addressee both accept it as true, and know that they both do so (Stalnaker 1978; Clark 1996: 93–96). The Common Ground presupposition of (1b) follows from its exclamative reading (cf. Zanuttini & Portner 2003). Example (2) shows that *that* is required in topicalized object clauses. Topics are necessarily discourse-old, which enables a parallel with (1): in both cases, we see that *that* is used to introduce information that is shared by Speaker and Addressee. Therefore, complementizer *that* is not interpretatively neutral; rather, the choice for *that* reflects the information status of the complement clause. Staps & Rooryck (2023) argue that it is not coincidental that a [+distal] element is used to mark reference to the Common Ground and other kinds of shared context. Rather, [+distal] *that* is used to mark a shared context with the Addressee, because she is seen as “far” from the Speaker.

In this article I propose a similar analysis for the Biblical Hebrew clausal connective *kī*, which I will show is also originally [+distal]. This particle occurs in a wide variety of environments. Previous scholars have struggled to account for this variety, both synchronically and diachronically. However, when we look at the data taking information

¹I use “clausal connective” as a neutral umbrella term for any grammatical item that connects two clauses, whether that is in a hypotactic/subordinating or paratactic/coordinating relationship.

status into account, we see that *kī* is chosen over alternatives when reference to the Common Ground is being made. This enables a primarily synchronic account which treats *kī* as a relatively neutral clausal connective, only presupposing that the information it introduces is in the Common Ground. The uses in different environments can be derived from this generic function given the right pragmatic cues.

This article makes the following contributions. First, it provides a new analysis of Biblical Hebrew *kī* with specific reference to its [+distal] origins, resolving the perceived high degree of polysemy. In doing so, it shows that paying attention to etymology is important to understand the use and distribution even of highly grammaticalized elements. Second, it provides support for the idea that a [+distal] element can be used to mark Common Ground by involving the Addressee, who is seen as “far” from the Speaker. This idea was first proposed by [Staps & Rooryck \(2023\)](#) on the basis of European languages, and supported with data from an unrelated language here. Furthermore, the Biblical Hebrew data are more complicated (and therefore provide stronger evidence), since *kī* can be used in a wider variety of contexts than, e.g., English *that*. Finally, by discussing the Biblical Hebrew data in depth, the article explores the different discursive effects that reference to Common Ground can have.

In the remainder of the introduction I briefly describe the Biblical Hebrew data pretheoretically (section 1.1), and provide more theoretical background on the proposed relationship between [+distal] and Common Ground (section 1.2). Section 1.3 provides necessary background on Biblical Hebrew. In section 2 I describe my method and give an overview of the data. The following sections discuss the different uses of *kī* in turn, showing how each use derives from the core function of marking Common Ground (sections 3 to 7). Section 8 concludes.

1.1 Biblical Hebrew *kī*

The Biblical Hebrew clausal connective *kī* can connect clauses with different relations to each other, and maps in translation onto a variety of English clausal connectives. Its use encompasses at least the introduction of object and subject clauses (‘that’, (3ab)), causal ‘because, for’ (3c), temporal/conditional ‘when’/‘if’ (3d), adversative ‘but’ (3e), resultative ‘so that’ (3f), and concessive ‘though’ (3g).

- (3) a. Gen. 16:4: וַיֵּרָא כִּי הָרָאָה
wat-t-ēre’ kī hārā-tā
 CONJ-F-see\PRET[3SG] COMP conceive\PFV-3F.SG
 ‘And she saw *that* she had conceived.’
- b. 2 Sam. 18:3c: טוֹב בִּי־תְהִי־לָנוּ מֵעִיר לְעֶזְרוֹ
tōb kī t-thaye lā=nū mē=’ir la=’zōr
 good COMP 2-be\IPFV[M.SG] for=us from=city for=help
 ‘It is better *that* you provide support for us from the city.”
- c. Gen. 34:14: לֹא נִכְלָ ... בִּי־חֲרָפָה הוּא לָנוּ
lā nīklā ... bi-ḥarāpā hūa lānū

lō' n-ūkal ... kī ḥerpā hū lā=nū
 not 1PL-be_able\IPFV ... COMP disgrace it for=us

'We cannot (do this, give our sister to a man who is uncircumcised), *for* it is a disgrace to us.'

- d. Gen. 32:18: וַיֹּצֵא אֶת־הָרְאשׁוֹן לְאֶמְרָא בִּי יַפְנֹשֶׁף עָשָׂו

wa-y-aṣaw 'et ḥā=rišōn lē='mōr kī
 CONJ-3M-command\PRET[SG] OBJ the=first to=say\INF COMP
y-ipāgāš=kā 'eśāw
 3M-meet\IPFV[SG]=you.OBJ Esau

'He (Jacob) commanded the first [servant], saying, "*If/When* Esau meets you, ..."

- e. 1 Sam. 12:12b: לֹא בִּי־מֶלֶךְ יִמְלֹךְ עָלֵינוּ

lō' kī melek y-implōk 'ālē=nū
 no COMP king 3M-rule\IPFV[SG] over=us
 'no; *but* a king shall rule over us!'

- f. Gen. 31:36: מַה חָטָאתִי בִּי דָלַקְתָּ אַחֵרַי:

mah ḥaṭṭā-tī kī dālaq-tā 'ahār=āy
 what sin\PFV-1SG COMP chase\PFV-2M.SG behind=me
 'How have I (Jacob) sinned *that* you (Laban) have chased after me?'

- g. 2 Sam. 12:12: כִּי אַתָּה עָשִׂיתָ בִּסְתֵר וְאֲנִי אֶעֱשֶׂה אֶת־הַדְּבָר הַזֶּה נֶגֶד כָּל־יִשְׂרָאֵל

kī 'attā 'āśī-tā b=as=sāter wa='ānī 'e'ēśe 'et had=dābār
 COMP you do\PFV-2M.SG in=the=secret and=I 1SG-do\IPFV OBJ the=thing
haz=ze neḡeḡed kāl yiśrā'el
 the=this before all.of Israel

'*Though* you have acted in secret, I will do this before all of Israel.'

In a broader typology of clause linkage based on parameters described by [Lehmann \(1988\)](#), *kī*-clauses behave most like "weakly desententialized noun clause[s] introduced by a universal subordinator", a type which also covers English *that*-clauses (e.g. *He believed [(that) I wrote a letter]*; [Lehmann 1988](#): 218). This assessment is based on various properties. First, *kī*-clauses can link clauses to clauses (in circumstantial uses) or to VPs/NPs (in complementizer uses), but there is no evidence for incorporation of *kī* into the matrix predicate and *kī*-clauses therefore retain a high degree of independence (e.g., in [12] below a prepositional phrase appears before *kī*). Second, unlike highly nominalized clauses (e.g. *her conceiving* instead of *she had conceived*; cf. [3a]), *kī*-clauses express tense and aspect and have relatively free word order (note the fronting of a prepositional phrase in [11] below).² The low degree of internal nominalization goes hand in hand with a lack

²Because volitive forms are frequently indistinguishable from regular prefix conjugation forms, it is hard to establish with certainty whether *kī*-clauses are restricted with respect to mood. Job 27:8 appears to be the only unambiguous volitive form in a *kī*-clause in the Hebrew Bible, but its reading is contested ([Dhorme 1967](#): 382–383). *Kī*-clauses are definitely restricted in terms of illocutionary force: though there are examples of imperatives and interrogatives in *kī*-clauses, they are used rhetorically and do

of distributional properties characteristic of NPs, such as adjectival modification, coordination with NPs, and selection by a large set of prepositions (in fact, only *‘al* ‘on’ and *‘ad* ‘until’ can select *kī*-clauses). Finally, *kī*-clauses cannot be “interlaced” with matrix clauses by sharing arguments or predicates (as is the case in, for instance, gapping and raising constructions).

Most scholarly discussion has focused not on structural and distributional properties of *kī* but on its functions.³ In particular, the large variety of contexts in which *kī* appears raises the question whether these different environments correspond to different functions of *kī* or whether these environments share certain properties that explain why *kī* is used in all of them. Scholarship in the past century or so has oscillated between describing *kī* as highly polysemous and highly abstract.⁴ Thus, whereas some have argued that all (or at least most) uses of *kī* can be reduced to a single semantic core (e.g. Muilenburg 1961; Schoors 1981), recent scholars have claimed that diachronic processes like grammaticalization have led to a highly polysemous lexeme (e.g. Locatell 2017, 2020).⁵

Regardless of these differences, the point of departure for both the synchronic and the diachronic approach is deixis. It is generally accepted that *kī* derives from an originally deictic morpheme **ka*, which has similar reflexes in at least Phoenician, Aramaic, and Arabic (Lipiński 2001: §49.9). Scholars in the synchronic tradition have generally tried to describe all uses as somehow deictic (e.g. Muilenburg 1961; Schoors 1981), whereas those with a primarily diachronic approach suggest possible grammaticalization paths from deictic origins (e.g. Locatell 2017).

What has not been given attention, unfortunately, is that **ka* is specifically a distal deictic morpheme, referring to things at some distance from the Speaker. This is especially clear in West Semitic, where **ka* also appears in demonstrative paradigms. According to

not have full illocutionary force: ‘Only fear Yahweh, for look at what great things he has done with you!’ (1 Sam. 12:24); ‘What will be done for the man who slays this Philistine ...? For who is this uncircumcised Philistine that he should taunt the armies of the living God?’ (1 Sam. 17:26).

³ A recent exception is Park (2016, 2023), who suggests that *kī* should be described as a “nominalizer”. Though the analysis in this article is compatible with this idea, my own assessment based on Lehmann’s (1988) parameters is different. Park (2015, 2016) suggests that both *kī* and *šer*, traditionally analyzed as a relativizer/complementizer, are nominalizers. This is based on a functional comparison to nominalizers such as Chinese *zhe*, which likewise form subject, object, relative, circumstantial, and matrix clauses. However, Park disregards distributional characteristics like the ones proposed by Lehmann (1988). For instance, unlike *kī* and *šer*, *zhe* can be modified by quantifiers and demonstratives (see also Holmstedt 2016: 89 n. 70). The problem is worst for *kī*-clauses, which are even less noun-like than *šer*-clauses, as can be seen from the smaller number of prepositions that can select them (Staps to appear). Furthermore, the grammaticalization path of *zhe* depends on the availability of pre-nominal modification (Yap & Wang 2011: 65), whereas modification is post-nominal in Biblical Hebrew. Because of this, the comparison does not go beyond functional similarities. Thus, while the functional overlap with Chinese *zhe* and other nominalizers remains interesting, it is unclear whether it can or needs to be explained with a unified synchronic analysis or shared grammaticalization path. Nevertheless, I stress that my analysis of *kī* is compatible with that of Park (2016, 2023).

⁴ See in particular Redslob (1835), Vriezen (1958), Muilenburg (1961), Schoors (1981), Bandstra (1982), Claassen (1983), Thorion (1984), Aejmelaeus (1986), Gross (1991), Benigni (1999), Follingstad (2001), Park (2016), and Locatell (2017, 2020). Some passages in works with a broader scope are relevant as well: Watts (1964: 118–149); Muraoka (1985: 158–164); Van der Merwe (1993: 38–41); and Conklin (2011: 46–59). I will not summarize related work here, as this has been done recently by Locatell (2017); instead, I only refer to these earlier sources where relevant to the argumentation of the present article.

⁵ Follingstad is an exception: he attempts to reduce *kī* to a “discourse deictic particle” with the function of setting up a new Mental Space and shifting viewpoint to this space. However, it is not clear that the Mental Space Theory is precise enough to exclude uses of *kī* that do not occur, and in some cases it actually makes incorrect predictions. For example, Follingstad (2001: 268–269) claims that *kī* can introduce a “hypothetical” conditional to which the Speaker does not need to commit, while the Speaker does need to commit to the truth of conditionals introduced by *ʔn* ‘if’. This is contrary to the consensus in the literature. In section 6.2 I show that the consensus can be maintained.

Hasselbach, **ka* “regularly marks far [i.e., distal] deixis in those languages in which it occurs” (Hasselbach 2007: 3).⁶ This observation provides a parallel with Germanic, where it is also the [+distal] form that grammaticalized into a clausal connective (i.e., English *that*). This [+distal] feature forms the basis for an interpretation based on Addressee involvement within the framework of Staps & Rooryck (2023), which I will introduce in section 1.2.

In short, the *empirical result* I present is that *kī* is used to introduce information that is in the Common Ground. My account is therefore primarily synchronic: roughly, *kī* is used when the information content of the complement clause is in the Common Ground, and the particular interpretation as causal, temporal, conditional, etc., largely depends on context.⁷ However, unlike previous authors with a synchronic approach, I also contrast *kī* with alternatives, to show that information that is not in the Common Ground is introduced in a different way.⁸ Under this analysis, only minor diachronic assumptions will be necessary. This is important, because any diachronic developments leading to the high number of uses in which *kī* can be used must have occurred before the appearance of written sources. Such diachronic developments can therefore be made conceivable on the basis of cross-linguistic parallels, but they cannot be tested on the Hebrew data themselves.

Based on the empirical results, the *theoretical analysis* I propose is that *kī* marks Common Ground because of an original [+distal] specification. This is based on previous work on European languages (primarily Germanic and Romance; Staps & Rooryck 2023), which is summarized in the next subsection.

1.2 Theoretical framework

The basic use of demonstrative *that* is to mark spatial distance (**this/that book over there*). However, in addition to this exophoric use, demonstratives have endophoric uses where they do not refer to a concrete entity in the physical world (Diessel 1999: 93–109). In these uses, other factors than spatial distance become relatively more important in the choice

⁶Lipiński (2001: §36.35, 36.37, 36.41) lists some demonstratives in Semitic and more remotely related Afro-Asiatic languages where **ka* appears to be used in proximal contexts, but these amount only to a handful of isolated instances, whereas the distal examples in West Semitic are more frequent and paradigmatic. In some languages where forms going back to **ka* are in paradigmatic contrast with the third person personal pronoun used as a [+distal] demonstrative, it appears that the forms based on **ka* are not distal but medial, referring to something near or known to the Addressee (e.g., ‘give me *that* (from **ka*)’ referring to an object near the Addressee; Jewish Babylonian Aramaic; Bar-Asher Siegal 2013: 82). This is compatible with my analysis, but a discussion of these demonstrative forms is out of scope here.

⁷One occasionally finds generic arguments against such a reductionist approach (Aejmelaeus 1986: 195; Locatell 2017: 114). However, the persistence of lexical meaning is, in fact, entirely expected in grammaticalization processes (Hopper 1991: 28–30). It is therefore not surprising if aspects of the [+distal] deictic meaning of Semitic **ka* are preserved in Biblical Hebrew *kī*, and my claim is that this is the most economical description of the data.

⁸This account aligns perhaps most with that of Park (2016), who likewise argues that different uses of *kī* are not different functions but arise from different contexts. However, the way in which this genericity arises is different. Park argues that *kī* is a nominalizer in the sense of Yap, Grunow-Hårsta & Wrona (2011), and points out that, cross-linguistically, nominalizations tend to occur in a wide variety of contexts. For me, the genericity arises because *kī* is a neutral clausal connective that only specifies that the information it introduces is in the Common Ground. Thus, the claim is more precise: the set of environments in which *kī* appears is exactly the set in which information in the Common Ground is introduced. This is compatible with, but not dependent on, Park’s nominalization analysis (see footnote 3); hence my use of the noncommittal umbrella term “clausal connective”.

between *this* and *that*. These include psychological factors, such as whether the referent is in joint attention or not (Peeters, Krahmer & Maes 2021). For instance, Colasanti & Wiltschko (2019) show that in several unrelated languages, proximal demonstratives are used to introduce entities that are novel to the Addressee (e.g. *this book I read*), whereas distal demonstratives are used to refer to familiar entities (e.g. *that book we talked about*).

In Staps & Rooryck (2023), this line of research is connected to one in which complementizer *that* is linked to Common Ground. In object clauses where complementizer *that* is optional, Storms (1966: 264) has suggested that the use of *that* (as opposed to zero) is used to “establish objective facts”, i.e., to set up Common Ground. Bolinger (1972: 56–60) links this additional objectivity to the original demonstrative character of *that*, noting that sentences with *that* (as opposed to zero) also more often refer back to already given information. Thus, emotion verbs are more felicitous when their complement clause contains some deictic element, like *so* or indeed complementizer *that*:

- (4) a. ?*I'm glad losses were light.*
 b. *I'm glad losses were so light.*
 c. *I'm glad that losses were light.* (Bolinger 1972: 57)

Exclamatives and topicalization are other contexts in which complementizer *that* can be linked to Common Ground status (see [1]–[2] above). Therefore, Staps & Rooryck (2023) propose that the [+distal] feature of the English demonstrative *that* is still present, but differently interpreted, when *that* is used as a complementizer. In particular, distal *that* can mark both “actual distance” and “Addressee involvement”: the first is most prevalent in exophoric demonstratives; the second in endophoric demonstratives and derived complementizers. The specific interpretation of Addressee involvement follows from the syntactic context: reference to familiar (as opposed to novel) entities in demonstrative *that*; reference to Common Ground and familiar information in complementizer *that*.⁹

Staps & Rooryck (2023) argue that it is the originally spatial [+distal] feature of *that* which leads to the notion of Addressee involvement. In their model, information content is positioned in an abstract space around Speaker and Addressee. The Common Ground is conceptualized as accessible and “close” to the Addressee. As a result, it is relatively “far” from the Speaker, for whom information content that is not shared with the Addressee is “closer”. Figure 1 (reproduced from Staps & Rooryck 2023: 7) clarifies the model. The circle around S stands for the information content tracked by the Speaker; the circle around A for the information content tracked by the Addressee. The intersection of these sets (shaded dark gray in fig. 1) can, under some assumptions, be seen as the Common Ground.¹⁰ The light gray shaded region represents the Speaker-private information content; information

⁹Staps & Rooryck (2023) introduce the term “Shared Discourse Space” for the region tracked by both the Speaker and the Addressee. Shared Discourse Space is more general than Common Ground and does not only contain information content. Since this article deals exclusively with the sentential domain, the more common term “Common Ground” suffices here. The argument is fully compatible with an analysis based on Shared Discourse Space, should this be needed for data not covered in this study.

¹⁰The main assumption is that this model is viewed from the perspective of the Speaker. Therefore, the circle around the Addressee represents not what is actually tracked by the Addressee, but what the Speaker *assumes to be* tracked by the Addressee. Without this assumption, it would not be possible for the Speaker to determine the content of the differently shaded regions,

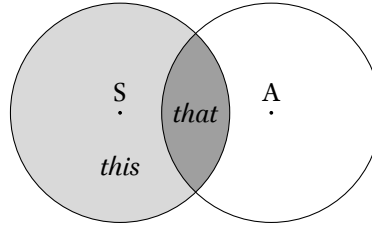


Figure 1: The information content tracked by the Speaker and Addressee. The intersection, the Common Ground, is seen as “far” from the Speaker (Staps & Rooryck 2023: 7).

that is not tracked by the Addressee. Even though the two gray regions are equally accessible to the Speaker, languages use [–distal] and [+distal] forms to refer to the light gray and the dark gray region, respectively. Thus, while the distance is measured from the Speaker’s origo, it reflects the accessibility to the Addressee, i.e., Common Ground status.

Speakers can interact with the abstract spatial model in fig. 1 in different ways. In the most basic cases, [+distal] forms are used to refer to discourse-old information content (which is in the Common Ground), as opposed to discourse-new information content. However, there are two cases in which discourse-new information content can be presented as part of the Common Ground: when the content can be *accommodated* by the Addressee and when it is *imposed* by the Speaker. Declarative sentences with discourse-new information content are typically modeled as a request or proposal to update the Common Ground (cf. Farkas & Bruce 2010: 92). However, by explicitly placing the content “near” the Addressee with a [+distal] element, the Speaker can present it as if it is already part of the Common Ground. The Speaker does so to signal that they expect that the request for a Common Ground update will be granted, or, alternatively, to emphasize that they do not allow the Addressee to reject the proposal to update the Common Ground. In the first case, the Speaker assumes that the Addressee can *accommodate* the information content; in the second case, the Speaker *imposes* the content on the Common Ground.¹¹

These three types of reference (to discourse-old, accommodated, and imposed information content) can all be seen as referencing to information conceptually “near” the Addressee. Discourse-old information content is “near” the Addressee because it is known and accessible to her. When new information content is placed “near” the Addressee by the Speaker, she can thereby suggest that the Addressee should easily be able to accommodate it. When this is not the case, this forces the Addressee to react; this is a case of information content imposed on the Common Ground. Since information content is seen as “near” the Addressee in all three types of reference, it is not surprising that an originally [+distal] deictic element like English *that* can be used to mark them: [+distal] *that* refers to a space “far” from the Speaker, but “near” the Addressee.

which is necessary if language use (e.g., the use of *kī*) depends on it. See Stalnaker (1978: 321), Clark (1996: 96), and Staps & Rooryck (2023: 1202–1203) for more discussion.

¹¹For more details and references, see Kocher (2022: 176–177). For examples, see the cases of accommodation below (e.g. [6; 17; 20; 23–24]) and compare with those of imposition (e.g. [11; 19; 25]).

1.3 Biblical Hebrew

Biblical Hebrew is a Northwest Semitic language spoken during the whole first millennium before the common era and into the first centuries of the common era. It is preserved primarily in the Hebrew Bible (which formed the basis for the Christian Old Testament), which is compiled from sources of diverse origins in terms of both space and time. Though working with an ancient language has the obvious drawback that no constructed examples can be tested with native speakers, this is mitigated by the availability of a long history of translation and interpretation. Thus, although the translations provided are my own, the underlying interpretations are corroborated by standard translations.

Like that of other Semitic languages, the Biblical Hebrew verbal system is based on triconsonantal roots, with templates consisting of affixes and vowel patterns to express Aktionsart (simple, pluractional, and causative) and Voice (active, passive, reflexive, and middle). When the meaning of a stem in a certain template is lexicalized I will simply give an appropriate English translation without glossing the template. There are two main finite conjugations: a suffix conjugation used in perfective/gnomic aspect and/or past tense contexts, and a prefix conjugation used in imperfective aspect, non-past tense, and various modal contexts. In keeping with traditional terminology I will gloss these forms PFV and IPFV, respectively, keeping in mind that their semantics is broader than this. In addition to these two conjugations there is a sequential preterite form used in narratives (traditionally called *wayyiqtol*, glossed here as CONJ-...-stem\PRET), and a sequential modal form (traditionally called *waqāṭaltí*, glossed here as CONJ-stem\MOD). Besides the imperative (IMP) of the second person, Biblical Hebrew has a jussive (JUSS) of the second and third person. Biblical Hebrew morphologically distinguishes two types of infinitives: the “infinitive construct” (INF), which is most like infinitives and gerunds in Indo-European; and an “infinitive absolute” (INFABS), the use of which is more complicated, but in the examples in this article is limited to strengthening an immediately following finite form of the same verb.

Nominals inflect for gender (masculine or feminine) and number (singular, plural, or dual; the latter of limited productivity). Verbs agree with their subject in person, gender, and number (but plural forms are used for dual subjects, and the first person has common gender). I only indicate gender on nouns where needed to clarify agreement. In possessive constructions, the possessed noun is in the “construct state”, which is morphologically distinguished from the default “absolute state” for certain combinations of gender and number; I gloss nouns in the construct state with “-of” (when morphologically marked) or “.of” (when syncretic with the absolute state).

2 Method and overview of the data

This study is based on an exhaustive analysis of the 808 uses of *kī* with a clausal complement in the narrative portions of the biblical books Genesis, Judges, Samuel, and Ruth. These books are considered to be similar in terms of time and place of origin, and thus

form a relatively homogeneous corpus.¹² I focus on narrative texts because these contain most direct speech, where we can expect the interaction with Common Ground to be the largest.¹³ However, I also included uses of *kī* outside direct speech. Therefore, in direct speech the terms “Speaker” and “Addressee” refer to story participants, but outside direct speech they refer to the narrator and reader, respectively.

Each instance of *kī* was classified as belonging to a certain use type (e.g. causal, complementizer, etc.).¹⁴ My classification of each instance can be found in the data set accompanying this article (Staps 2023). The use types are based on the common categories found in reference works and literature on *kī*: (a) introducing object and subject clauses (‘that’; tagged as “complementizer”), (b) causal ‘because, for’, (c) adversative ‘but’, (d) causal-adversative ‘not X, *because/but rather* Y’, (e) conditional ‘if’, (f) temporal ‘when’, (g) resultative ‘so that’, and (h) concessive ‘though’.¹⁵ Instances where the *kī*-clause does not seem to relate to a corresponding “main” clause were classified as (i) standalone; this group will be further subcategorized in section 7. Five cases were ambiguous; I ignore them here for ease of exposition.¹⁶

Each instance was also tagged for the way it interacts with the Common Ground. As explained in section 1.2, the information content of the *kī*-clause can be (a) discourse-old and thus part of the Common Ground, (b) easily accommodated by the Addressee as new Common Ground, or (c) imposed on the Common Ground by the Speaker for discursive effect. There are some cases where the information content in the *kī*-clause does not fit either of these cases; these were classified as (d) rest. I will discuss these separately below, to show why they do not constitute counter-examples for my claim that *kī* marks reference to Common Ground. Nevertheless I also include many examples where information is in the Common Ground, in order to demonstrate the ways in which this can be used in discourse. Figure 2 shows the distribution of types of reference to the Common Ground for each use type of *kī*; the numbers from which this graph has been compiled are provided in table 1.

It can immediately be seen that in the vast majority of cases, the information content of the *kī*-clause is in the Common Ground, easily accommodated, or imposed: in only

¹²Biblical Hebrew texts are usually divided into Early and Late Biblical Hebrew according to whether they are written before or after the Judean exile in 586 BCE (Young & Rezetko 2008: 7). Genesis, Judges, and Samuel are generally dated as pre-exilic (Young & Rezetko 2008: 10–11). Certain passages that are frequently dated to an even earlier stage (Archaic Biblical Hebrew: Genesis 49; Judges 5; 2 Samuel 22) are excluded here because they are poetic. Ruth is generally considered to have been written in the post-exilic period using an archaizing style (Young & Rezetko 2008: 99, 138). This makes it somewhat of an edge case, but the 28 instances of *kī* in Ruth follow the same patterns as those in Genesis, Judges, and Samuel. All examples in this article come from uncontested pre-exilic books.

¹³It is quite conceivable that the use of *kī* in poetry follows a different, but comparable, distribution (see e.g. Meyer 2001). In poetry it is often much less clear what the Common Ground contains, so the Common Ground may be a weaker factor in choosing between *kī* and alternatives in poetic texts.

¹⁴I excluded instances of the fossilized construction *כִּי יִמ* *kī ’im*, and two instances of bare *kī*, in the meaning ‘except’, which I assume to have grammaticalized more or less independently (pace Park 2023; see Staps to appear).

¹⁵Naturally, there are cases that can be classified as one of two categories, in particular in the temporal/causal, temporal/conditional, and causal/resultative categories (cf. Locatelli 2020). My argument does not rely on a sharp distinction between these categories, so I have in these cases selected what seemed to be the best-fitting category without spending too much thought on it.

¹⁶Gen. 8:21 (causal/concessive); 21:7 (adversative/standalone); 38:16c (conditional/resultative); 1 Sam. 15:24a (resultative/standalone); 2 Sam. 18:3b (causal *or* adversative, but not causal-adversative). See the data set for more details.

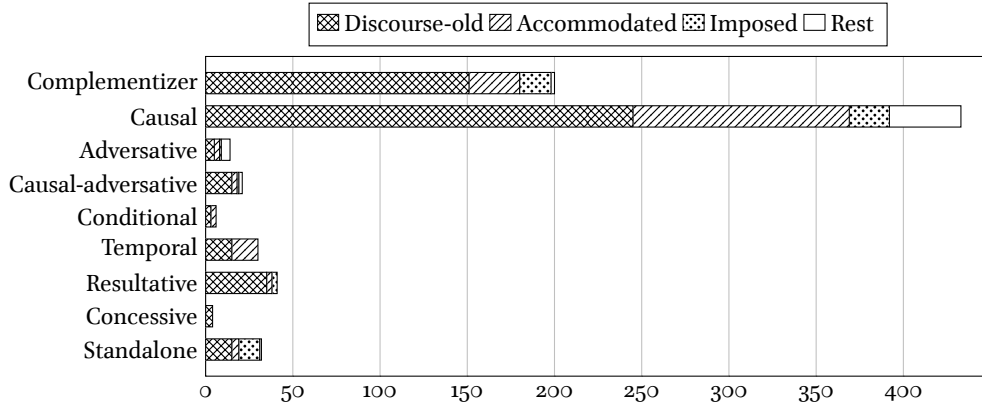


Figure 2: Types of reference to the Common Ground for each use type of $k\bar{i}$.

Type	Common Ground		Accommodated		Imposed		Rest		Total
Complementizer	151	76%	29	15%	18	9%	2	1%	200
Causal	245	57%	124	29%	23	5%	41	9%	433
Adversative	5	36%	3	21%	1	7%	5	36%	14
Causal-adversative	15	71%	3	14%	1	5%	2	10%	21
Conditional	3	50%	3	50%	0		0		6
Temporal	15	50%	15	50%	0		0		30
Resultative	35	85%	3	7%	3	7%	0		41
Concessive	4	100%	0		0		0		4
Standalone	15	47%	4	13%	12	38%	1	3%	32
Total	488	62%	184	24%	58	7%	51	7%	781

Table 1: Distribution of instances of $k\bar{i}$ and the use of Common Ground per type.

7% of the total number of cases is there no reference to the Common Ground. The cases where $k\bar{i}$ apparently does not interact with the Common Ground are almost exclusively isolated in a few use types (causal, adversative, and causal-adversative).

The following sections discuss each type of $k\bar{i}$ -clause in turn, beginning with object and subject clauses (section 3), followed by several types of circumstantial clauses (sections 4 to 6), and finishing with standalone $k\bar{i}$ -clauses (section 7).

3 Object and subject clauses introduced with $k\bar{i}$

It is cross-linguistically not uncommon for complementizers such as $k\bar{i}$ to develop from deictic elements.¹⁷ I will not concern myself here with the question why an originally

¹⁷As already mentioned above, English *that* is both a distal demonstrative (*that book*) and a complementizer (*I know that ...*); it shares these functions with Semitic **ka*. In Latin, the complementizer *quod* (*Scio quod ...* ‘I know that ...’) is made up of the interrogative element *qu-* and the medial demonstrative *-id*. In Russian, the complementizer *što* (*ya znayu što ...* ‘I know that ...’) is related to the demonstratives *eto* (*eto dom* ‘this house’) and *to* (*von to pal'to* ‘that coat over there’). For references on

[+distal] deictic element becomes a complementizer; this has been addressed in great detail elsewhere.¹⁸ Rather, I will compare complement clauses introduced by *kī* with other complementation strategies to show that *kī*-clauses are used specifically when reference to the Common Ground is being made.

The complement clauses introduced by *kī* in my corpus can be divided into four main categories depending on the type of matrix predicate. By far the most common are (a) verbs of perception (רָאָה *rāʾā* ‘see’; שָׁמַע *šāma* ‘hear’) and (b) cognitive verbs (יָדָע *yāda* ‘know’); less common are (c) speech verbs (הִגִּיד *higgīd* ‘inform’) (see Miller 2003: 98 on the relative rarity of this category). The remaining category contains (d) miscellaneous constructions: cases where the object clause is governed by a noun (עֵד *ʾēd* ‘witness’) or preposition (עַד *ʾad* ‘until’), as well as cases where *kī* introduces a subject clause.

3.1 Verbs of perception

The verb רָאָה *rāʾā* ‘see’ occurs 65 times with *kī* in my corpus, שָׁמַע *šāma* ‘hear’ 20 times, and הִתְבַּשֵּׂר *hitbaššēr* ‘receive good news’ once. These verbs are most frequently used to describe an event in which the subject becomes aware of information that was already known to the Addressee (cf. English *see* in the sense ‘realize’). Thus, in (5), even though the information in the *kī*-clause is novel for the matrix subject (Hagar), it is clearly in the Common Ground for the Addressee (the reader), based on the immediately preceding context.

- (5) Gen. 16:4: וַיָּבֹא אֶל־הָגָר וַתֵּהָר וַתֵּרָא כִּי הָרְתָה
way-y-ābōʾ ʾel hāgār wat-t-ahar
 CONJ-3M-come\PRET[SG] to Hagar CONJ-F-conceive\PRET[3SG]
wat-t-ēreʾ kī hārā-tā
 CONJ-F-see\PRET[3SG] COMP conceive\PFV-3F.SG
 ‘And he came into Hagar and she conceived, and she saw *that* she had conceived.’

The information can also be accommodated, as in (6). In this example, the fact that the man cannot overpower Jacob is easily accommodated by the Addressee (the reader) given the information that they wrestle for a long time (until daybreak).

the grammaticalization process in Indo-European languages, see Kuteva et al. (2019: 134–135), who also provide some examples outside Indo-European: Hopper & Traugott (2003: 185) discuss Kunwinjku (Gunwinyguan) *gunu* ‘that’ and Lehmann (2015: 65) mentions but does not discuss Akkadian (Semitic) *ša* ‘COMP’ (< *šu* ‘he, that’) and Nahuatl (Uto-Aztecan) *in*.

¹⁸It is usually assumed that the English complementizer *that* developed from a cataphoric pronoun: *Galileo said that_i: [the earth is round]_i* > *Galileo said [CP that the earth is round]* (Roberts & Roussou 2003: 113, and references therein). This grammaticalization path is somewhat problematic for Hebrew *kī*, which never was a demonstrative. However, this grammaticalization path has been challenged in recent work (Kayne 2014: 189; Axel-Tober 2017; though see also the critique of Kayne’s analysis by de Cuba 2023), for example because it does not explain why complementizers are typically based on [+distal] elements (**Galileo said this the earth is round*). For this reason, I assume that complementizers are not merely reanalyzed demonstratives, but carry the same deictic [+distal] feature in a different syntactic environment; see Staps & Rooryck (2023).

- (6) Gen. 32:25–26: וַיִּאָבֶק אִישׁ עִמּוֹ עַד עֹלֹת הַשָּׁחַר: וַיֵּרָא כִּי לֹא יָכֹל לוֹ
 way-y-ē^ābēq iš imm=ō ad ʾālōt haš=šāhar
 CONJ-3M-wrestle\PRET[SG] man with=him until go_up\INF.of the=daybreak
 way-y-ar' kī lō' yākōl l=ō
 CONJ-3M-see\PRET[SG] COMP not be_able\PFV[3M.SG] to=him
 'And a man wrestled with him until daybreak, and he realized *that* he could not overpower him.'

A comparison with other complementation strategies is most helpful to show that *kī*-clauses are associated with Common Ground; I will use רָאָה *rāʾā* 'see' as a running example. In my corpus, this verb also occurs with clausal complements introduced by וַהֲגִיד *wa-hinnē* 'and behold'.¹⁹ As pointed out by a reviewer, this latter strategy is not syntactic but pragmatic in nature, but this does not preclude a comparison between the two. With *wa-hinnē*, the information given in the complement clause is new and not anticipated by the Addressee. In (7), the Addressee is the reader:²⁰

- (7) Gen. 22:13: וַיֵּשׂ אַבְרָהָם אֶת־עֵינָיו וַיֵּרָא וַהֲגִיד־אֵיל אֲחֵר נֹאֲחַז בְּסֶבֶךְ בְּקִרְנָיו
 way-y-iššā' ʾabrāhām ʾet ʿen=āyw way-y-ar'
 CONJ-3M-lift\PRET[SG] Abraham OBJ eye=DU.his CONJ-3M-see\PRET[SG]
 wa=hinnē ʾayil ʾaḥar neʾšḥaz b=as=səbak bə=qarn=āyw
 and=behold ram behind hold\MID.PFV[3M.SG] in=the=bush in=horn=PL.its
 'As Abraham looked up, he saw—and *behold!*—a ram behind [him]; it had been caught with its horns in a bush.'

An event being seen can also be described with a nominal complement, when modified by a circumstantial qualifier headed by a participle ([8]; cf. Isaksson 2009: 57–59). This strategy can be combined with *hinnē* 'behold', in which case the information is noteworthy (9). When *hinnē* 'behold' is not used (as in [8]), the information is new and unexpected (hence not easily accommodated), but also not marked as particularly noteworthy or immediately requiring the Addressee's attention; this strategy thus provides a middle ground between complementation with *kī* on the one hand, and *hinnē* on the other.

- (8) 2 Sam. 11:2: וַיֵּרָא אִשָּׁה רֹחֶצֶת מֵעַל הַגָּג
 way-y-ar' iššā rōḥeš-et mē=ʾal hag=gāg
 CONJ-3M-see\PRET[SG] woman bathe\PTCP-F.SG from=on the=roof
 '... and he saw a woman bathing from upon the roof.'

¹⁹There are some cases where the complement is introduced by an interrogative pronoun. In these cases the information content is evidently not known to the Speaker (e.g. Gen. 37:20: 'and we'll see *what* will become of his dreams!'), so I will not compare these instances to complementation with *kī* here.

²⁰Also Gen. 19:28; Jdg. 21:21. With a participle the complement can be either clausal or nominal: Gen. 18:2; 24:63; 26:8; 29:2; 37:25; Jdg. 3:24; 9:43; 1 Sam. 10:11; 2 Sam. 13:34. Two examples are ambiguous between participle and suffix conjugation (PFV): Gen. 33:1; 2 Sam. 18:24.

- (9) Gen. 18:2: וַיִּשָּׂא עֵינָיו וַיַּרְא וְהִנֵּה שְׁלֹשָׁה אָנָשׁ נֹצְבִים עָלָיו
 way-y-iššā' ēn=āyw way-y-ar' wə=hinnē šəlōšā
 CONJ-3M-lift\PRET[SG] eye=DU.his CONJ-3M-see\PRET[SG] and=behold three
 ānāš-īm niššāb-īm āl=āyw
 man-PL stand\PTCP-M.PL on=him
 'He lifted up his eyes and he saw—and behold!—three men were standing in front of him.'

Hinnē 'behold' also occurs with *kī*, but this combination is highly marked, because it suggests both noteworthy and discourse-old information status. In my corpus it occurs only once (10).²¹ Here, an angel tries to convince a woman who is barren that she will become pregnant. The angel has already said this once before, justifying the use of *kī*, but it remains a noteworthy fact that the woman will not have gotten used to yet, justifying the use of *hinnē*.

- (10) Jdg. 13:5: כִּי הִנֵּךְ הָרָה וַיִּלְדֶּת בֵּן
 kī hinn=āk hārā wə-yōlad-t bēn
 COMP behold=you pregnant CONJ-bear\MOD-2F.SG son
 '(Behold, you are barren ... but you shall conceive ... So be careful ...) for behold, you are pregnant and shall bear a son.'

The overall division of labor is clear: *hinnē* 'behold' introduces discourse-new or noteworthy information, and *kī* introduces discourse-old or easily accommodated information. A nominal complement modified by an attributively used participle, unless combined with *hinnē* 'behold', is used for information that is new and not expected, but also not particularly noteworthy.

3.2 Cognitive verbs

The class of cognitive verbs consists primarily of יָדָע *yāda* 'know' (72 times); other verbs in this class are נִחַם *niḥam* 'regret' (4 times), בִּין *bīn* 'understand' (twice), כִּיֵּהַד *kihēd* 'hide', and זָכַר *zākar* 'remember' (both once). All these less frequent predicates are factive and thus necessarily refer to Common Ground (e.g. 1 Sam. 15:35b: 'I have come to regret (*niḥam*) that I have made Saul king').

²¹ Outside my corpus, *kī hinnē* only appears in prophetic texts: Isa. 3:1; 26:21; 60:2; 65:17, 18; 66:15; Jer. 1:15; 8:17; 25:29; 30:3, 10; 45:5; 46:27; 49:15; 50:9; Ezek. 30:9;]bref369.

The verb *yāda* ‘know’ is often used in the same sense as *rāʾ* ‘see’, meaning ‘realize’, and then has a complement that is obviously discourse-old or accommodated by the Addressee. However, in my data set, *yāda* ‘know’ is used more frequently than *rāʾ* to impose information on the Common Ground, as in the following example:²²

- (11) 1 Sam. 28:1 תִּדַּע כִּי אִתִּי תֵצֵא בַּמִּלְחָמָה
t-ēda *kī* *’itt=ī* *t-ēṣe* *b=am=maḥāne*
 2-know\IPFV[M.SG] **COMP** with=me 2-go_out\IPFV[M.SG] in=the=camp
 ‘(And Achish said to David:) “You should know ... *that* you will go out with me in battle.”’

Here, David, an Israelite, has sought refuge with the Philistine king Achish. When the Philistines prepare to fight Israel, Achish makes sure that David understands that he has to fight on Achish’s side now, and cannot refuse to participate. This is not something they discussed before. *Kī* imposes this information on the Common Ground, which works well for a command. Apart from such cases of imposition, however, *kī*-clauses with cognitive verbs are quite similar to those with verbs of perception.

3.3 Speech verbs

The most frequent speech verb with *kī*-clauses is *higgīd* ‘inform’ (13 times, and 5 times in the passive); we also have *nišba* ‘swear’ (2 times), and *āmar* ‘say’, *biššēr* ‘bring good news’, and *gālā* ‘uncover the ear’ (1 time each).²³ It is noteworthy that *āmar* ‘say’, which is overall more frequent, is used so little with *kī*. This suggests a contrast between this verb and *higgīd* ‘inform’, wherein the latter is preferred when the complement is in the Common Ground. Thus, in the following example, the indirect speech complement repeats content that was already introduced to the reader in the previous sentence, and accordingly, *higgīd* ‘inform’ is used with *kī*.²⁴

- (12) Jdg. 9:46–47: וַיָּבֹאוּ אֶל-צִיְרִיחַ בֵּית אֵל בְּרִית: וַיֹּגֵד לְאַבִּימֶלֶךְ כִּי הִתְקַבְּצוּ כָּל-בְּעָלֵי מְגִד־שָׂכָם:
way-y-ābō-ū *’el* *ṣarīaḥ* *bēt* *’el*
 CONJ-3M-come\PRET-PL to stronghold.of house.of El
bərīt *way-y-uggad* *la=’ābīmelek* *kī*
 Berith CONJ-3M-inform\PASS.PRET[SG] to=Abimelech **COMP**
hītə-qabbəṣ-ū *kāl* *ba’āl-ē* *mīgədal šəkem*
 REFL-gather\PLURACT.PFV-3PL all.of lord-PL.of tower.of Shechem

²²Specifically, with *rāʾ* ‘see’, 45 examples refer to the Common Ground, of 15 the content can be accommodated, of 1 the content is imposed, and 2 are exceptional. With *yāda* ‘know’, 53 examples refer to the Common Ground, of 4 the content can be accommodated, of 13 the content is imposed, and none are exceptional.

²³I follow Miller (2003: 103–116) in rejecting the idea, put forward by Budde (1889), Goldenberg (1991), and others, that *kī* can introduce direct speech: *kī* either introduces indirect speech, or it is the first word in the reported direct speech. I discuss cases of the former here; when *kī* is part of the reported speech it can have any other function, which I discuss throughout the rest of this article.

²⁴Keep in mind here, again, that the Addressee, for our purposes, is not the Addressee of the speech event (the one who is being informed, here Abimelech), but the Addressee of the reporting of the speech event (i.e., the reader of the text).

‘And [the leaders of the Tower of Shechem] came to the stronghold of the house of El-Berith. And Abimelech was informed *that* all the leaders of the tower of Shechem had gathered.’

A comparison with asyndetic indirect speech (Miller 2003: 119–123) is interesting here.²⁵ It should be noted that asyndetic indirect speech only occurs embedded within direct speech (Miller 2003: 120), and the absence of a complementizer may therefore be attributed in part to register (cf. the more frequent omission of the English complementizer *that* in direct speech: Elness 1984; Rissanen 1991). However, when we compare indirect speech with and without complementizer, both embedded in direct speech, a Common Ground effect can be observed:

- (13) Gen. 12:13: אֶמְרִינָא אֶתְּחִי אֵשֶׁר
 ’imr-ī nā’ āhōt=ī ātt
 say\IMP-F.SG please sister=mine you
 ‘(Abram said to his wife Sarai: “...) Please say Ø you are my sister.”’
- (14) Gen. 12:18: וַיִּקְרָא פַרְעֹה לְאַבְרָם וַיֹּאמֶר ... לָמָּה לֹא הַגַּדְתָּ לִּי כִּי אִשְׁתִּי הִוא
 way-y-iqrā’ p̄ar’ōh lə=’abrām way-y-ō’mer ... lāmmā
 CONJ-3M-call\PRET[SG] Pharaoh to=Abram CONJ-3M-say\PRET[SG] ... why
 lō’ higgad-tā l=ī kī išt=akā hū
 not inform\PFV-2M.SG to=me COMP wife=yours she
 ‘So Pharaoh called Abram and said: “... Why did you not tell me *that* she is your wife?”’

In (13), Abram’s idea to pretend that Sarai is his sister is new to the Addressee (Sarai), and there is no reason why it should be easily accommodated. The analysis proposed here correctly predicts that *kī* is not used. However, in (14), Pharaoh has found out that Sarai is Abram’s sister. The information is in his and Abram’s Common Ground, and the indirect speech is introduced with *kī* accordingly.²⁶

One could object that the pair in (13–14) is not minimal because two different verbs are used: *higgad* ‘inform’ with *kī* in (14) and *amar* ‘say’ without *kī* in (13). Since asyndetic indirect speech complements are only ever found with *amar* ‘say’ (Miller 2003: 121), a better minimal pair cannot be given.²⁷ However, note that the simple fact that *kī* appears with one verb and not the other can and should be seen as reflecting the function of the complementizer to mark Common Ground: meaning ‘inform’, *higgad* is simply much more suitable to talk about common knowledge, whereas *amar* ‘say’ is more often used

²⁵Indirect speech can also be introduced with the relative pronoun *šer* (Miller 2003: 97–98), but since this is rare and mostly a feature of Late Biblical Hebrew, I will not compare the two here.

²⁶In Gen. 12:19, Pharaoh continues to ask: ‘why did you say, “she is my sister”?’ without *kī*. But *kī* is excluded here since the complement is a direct speech report, as can be seen from the pronoun shift (Miller 2003: 120).

²⁷The one example of *amar* ‘say’ with a *kī*-clause has the verb in the meaning ‘think, say to oneself’: ‘I *thought that* you really hated [your bride], so I gave her to your best man’ (Jdg. 15:2). Common Ground is imposed here; the Speaker informs the Addressee of an assumption he made based on the Addressee’s earlier behavior.

in contexts where the Addressee is given new information, such as imperatives (13). This distribution therefore confirms the hypothesis concerning *kī*.

3.4 Miscellaneous complementation structures

Although complementizer *kī*-clauses are usually object clauses complementing verbs, they can also complement nouns and prepositions, or function as subject clauses. With nouns we only find *ʿēd* ‘witness’ (1 Sam. 12:5; Ruth 4:9) and *ʿēdā* ‘legal proof’ (Gen. 21:30b); with prepositions we only have *ʿad* ‘until’ (3 times).

The nouns *ʿēd* ‘witness’ and *ʿēdā* ‘legal proof’ are used with *kī* only to establish who can corroborate a certain fact. This is always a known fact, that is, a fact in the Common Ground:

- (15) 1 Sam. 12:5: וַיֹּאמֶר אֵלֵיהֶם עַד יְהוָה בָּכֶם ... כִּי לֹא מִצְאָתֶם בְּיָדִי מֵאוֹמָה
way-y-ō'mer ʾālē=hem ʿēd yhw h bā=kem ... kī lō'
 CONJ-3M-say\PRET[SG] to=them witness Yahweh against=you ... COMP not
məṣā'-tem bə=yā'd=ī mə'umā
 find\PFV-2M.PL in=hand=mine anything
 ‘(And they said: “You have not wronged us ...”) So he said to them: “Yahweh is a witness against you ... *that* you have not found any [charge] against me.”’

Here, the complement clause simply reiterates what has already been said in the previous verse. Without *kī*, *ʿēd* ‘witness’ can be used with new information. In the following example, it is clear that Laban and Jacob are going to form some kind of covenant, but the preceding context provides no information based on which the Addressee can deduce that the cairn will be a marker of the border between them. Therefore *kī* would be inappropriate:

- (16) Gen. 31:52: עַד הַגֹּל הַזֶּה ... אִם-אֶעֱבֹר אֵלַיְךָ אֶת-הַגֹּל הַזֶּה וְאִם-אֶתָּה לֹא-תֵעָבֵר אֵלַי אֶת-הַגֹּל
ʿēd hag=gal haz=zε ... ʾim ʾānī lō' ʿ-εʿbōr ʿelε=kā ʿet
 witness the=cairn the=this ... if I not 1SG-pass_over\IPFV to=you OBJ
hag=gal haz=zε wə=ʾim ʾattā lō' t-aʿbōr ʿela=y ʿet
 the=cairn the=this and-if you not 2-pass_over\IPFV[M.SG] to=me OBJ
hag=gal haz=zε ... lə=rāʿā
 the=cairn the=this ... for=evil
 (Laban to Jacob:) “This cairn is a witness ...: Ø I will not pass this cairn to you, and you will not pass this cairn ... to me to do harm.”

Each instance with *ʿad* ‘until’ describes a direct consequence of the matrix clause (Gen. 26:13; 41:49; 2 Sam. 23:10). For example, in (17), being very wealthy is a direct consequence of becoming more and more wealthy. The *kī*-clause is therefore easily accommodated in the Common Ground:

- (17) Gen. 26:13: וַיֵּגְדַל הָאִישׁ וַיֵּלֶךְ הָלוֹךְ וַיֵּגְדַל עַד כִּי־יֵגְדַל מְאֹד׃
way-y-iḡdal hā=’iš way-y-ēlek hālōk
 CONJ-3M-be_great\PRET[SG] the=man CONJ-3M-go\PRET[SG] go\INFABS
wə=ḡāḏēl ‘ad kī ḡāḏal mə’ōd
 and=be_great\PTCP[M.SG] until **COMP** be_great\PFV[3M.SG] very
 ‘The man was wealthy, and he became more and more wealthy²⁸ to the point *that* he was very wealthy.’

The remaining cases are subject clauses. Two of these begin with *אִם* *ap kī* ‘[it is] even [the case] that’ (Gen. 3:1; 1 Sam. 14:30). I first discuss the other four, which are more straightforward: they all nominalize a previously introduced proposition, and thus refer to the Common Ground (1 Sam. 25:30; 2 Sam. 9:1; 18:3; Ruth 2:22). For instance, in (18), the proposition in the *kī*-clause is implied by the earlier ‘you will not go out’:

- (18) 2 Sam. 18:3c: לֹא תֵצֵא ... וְעַתָּה טוֹב כִּי־תִהְיֶה־לָּנוּ מִעִיר לְעֹזֶר׃
lō’ t-ēṣē’ ... wə=’attā tōb kī t-ihaye lā=nū
 not 2-go_out\IPFV[M.SG] ... and=now good **COMP** 2-be\IPFV[M.SG] for=us
mē=’ir la=’zōr
 from=city for=help
 ‘(The king said to the people: “I myself will also go out with you.” But the people said:) “You will not go out ...; now, it is better *that* you provide support for us from the city.”’

These cases thus confirm the hypothesis that *kī* marks the use of Common Ground (unfortunately, however, they cannot be compared to asyndetic finite subject clauses; these do not exist).

The cases with *אִם* *ap kī* ‘[it is] even [the case] that ...’ are as follows:²⁹

- (19) Gen. 3:1: אָמַר אֱלֹהִים לֹא תֹאכְלוּ מִכָּל־עֵץ הַגָּן׃
‘ap kī ‘amar ‘ēlōhīm lō’ t-ō’kəl-ū mik=kōl ‘ēṣ
 even **COMP** say\PFV[3M.SG] God not 2-eat\IPFV-M.PL from=all.of tree.of
hag=ḡān
 the=garden
 ‘(And the snake said to the woman:) “Is it *really* the case *that* God has said: ‘You shall not eat from any tree of the garden?’”’

²⁸Reading *ḡāḏōl* (INFABS) for *ḡāḏēl*, following the leading scholarly edition (Elliger & Rudolph 1997); for the durative interpretation of the infinitive absolute, see Gzella (2008).

²⁹The construction *ap kī* can sometimes be read as ‘how much more/less’, introducing a clause that provides a stronger reason for an implicit assertion than the reason provided in the previous clause (Van der Merwe, Naudé & Kroeze 2017: §40.14.1b). Three cases where *ap kī* should be read together are classified as causal (1 Sam. 21:6b; 23:3; 2 Sam. 16:11a).

(20) 1 Sam. 14:30a: אֶף כִּי לֹא אָכַל אֶכְל הַיּוֹם הַזֶּה מִשְׁלַל אִיְבֹיו אֲשֶׁר מָצָא כִּי עָתָה לֹא־רִבְתָּה מִכָּה בַּפְּלִשְׁתִּים:

'ap kī lū' ākōl ākal hayyōm hā=ām
 even COMP had_not eat\INFABS eat\PFV[3M.SG] today the=army(M)
 miš=šalal ōyab=āyw āšer māšā' kī 'attā lō'
 from=provision.of enemy=PL.its REL find\PFV[3M.SG] COMP now not
 rābā-tā makkā b=ap=pāliš-ti-m
 be_great\PFV-3F.SG slaughter(F) against=the=Philistine-PL

‘(See how my eyes gleamed when I tasted just a little of this honey.) It’s *certainly* the case *that*, had the army today eaten from the enemies’ provision which it found, that now the slaughter of the Philistines would have been greater.’

Example (19) is a case of information imposed on the Common Ground: the snake presumably knows that God has only forbidden the people to eat from one tree, but gains the woman’s trust by pretending he is asking a simple question. By pretending that he is ill-informed, the snake presents itself as harmless to the woman, which it will subsequently exploit. In (20), the information is accommodated in the Common Ground. In the previous clause, the Speaker has suggested that he was strengthened by eating just a little of the enemies’ provisions; the following verse simply extends this to the rest of the army.

3.5 Summary

In conclusion, both the distribution of complementizer *kī* over different matrix predicates and a comparison with other complementation strategies support the hypothesis that *kī* marks information in the Common Ground, or information that is easily accommodated in the Common Ground. In terms of distribution we may note the frequent use with רָאָה *rā’ā* in the sense ‘realize (old information)’ rather than the literal ‘see (something new)’, as well as the preference for הִגִּיד *higgīd* ‘inform (of old information)’ over אָמַר *amar* ‘say (something new)’. I have compared *kī* to various other complementation strategies, such as וַיַּהֲרֹא *wa-hinnē* ‘and behold’ and asyndetic indirect speech, which can all be shown to be used when the complement is not in the Common Ground yet, in contrast to the cases with *kī*. The following sections proceed with the analysis of circumstantial *kī*-clauses.

4 Causal *kī*

As mentioned in section 2, *kī* most frequently introduces a causal clause, which gives the cause, reason, or ground for the event described in the main clause.³⁰ In the majority of cases (almost 90%), the cause given in the clause is either already in the Common Ground or easily accommodated by the Addressee. It is not uncommon for causal conjunctions to be sensitive to reference to Common Ground (see e.g. Dancygier & Sweetser 2000 on

³⁰The discussion of causal-adversative *kī* is delayed until section 5, where adversative *kī* is discussed as well.

English *since*). Examples of such straightforward cases of causal *kī* are provided in (21–24). In (21), the reason is in the Common Ground, because it is cultural knowledge which the Speaker (writer) assumes the Addressee (reader) to have. In (22), the reason is in the Common Ground because it has been given in the previous clause.

- (21) Gen. 34:14: לֹא נוּכָל ... בְּיַחְרָפָהּ הוּא לָנוּ:
lō' n-ūkal ... kī ḥerpā hī lā=nū
 not 1PL-be_able\IPFV ... COMP disgrace it for=us
 'We cannot (do this, give our sister to a man who is uncircumcised), *for* it is a disgrace to us.'

- (22) Jdg. 14:17: וַתִּבְדֹּק עָלָיו שִׁבְעַת הַיָּמִים אֲשֶׁר־הָיָה לָהֶם הַמְּשָׁתָה וַיְהִי | בַּיּוֹם הַשְּׁבִיעִי וַיַּגֵּד־לָהּ כִּי הִצִּילָהּהוּ
wat-t-ēbk āl=āyw šib'a-t hay=yām-īm āšer hāyā
 CONJ-F-cry\PRET[3SG] on=him seven-of the=day-PL REL be\PFV[3M.SG]
lā=hem ham=mište wa-y-ahī b=ay=yōm haš=šabiī
 for=them the=party(M) CONJ-3M-be\PRET[SG] on=the=day the=seventh
way-y-aggēd l=āh kī hēšiq-at=hū
 CONJ-3M-inform\PRET[SG] to=her COMP press\PFV-3F.SG=him
 'She (Delilah) cried in his (Samson's) presence during the seven days of their party, and on the seventh day he told her, *because* she pressed him so.'

In (23), it is said that the grain that Joseph is storing is 'a very great quantity', and eventually he has to stop counting it. Based on this, the information in the *kī*-clause (that the stored grain had become immeasurable) is easily accommodated. Similarly, in (24), Samuel has to take oil and go to Jesse. Based on this (and the fact that Saul has been rejected as a king in the preceding chapter), it is easily accommodated that someone in Jesse's family will be the new king.

- (23) Gen. 41:49b: וַיִּצְבֹּר יוֹסֵף בָּר כְּחֹל הַיָּם הַרְבֵּה מְאֹד עַד בְּיַחְדָּל לְסַפֵּר כִּי־אֵין מִסְפָּר:
way-y-išbōr yōsēp bār ka=ḥōl hay=yām harbe
 CONJ-3M-store\PRET[SG] Joseph grain like=sand.of the=sea be_great\CAUS.INF
mā'ōd 'ad kī ḥādal li=spōr kī 'en mišpār
 very until COMP stop\PFV[3M.SG] to=count\INF COMP not_exist number
 'And Joseph stored grain—as much as the sand of the sea, a very great quantity—to the point that he stopped counting it *because* it was immeasurable.'

- (24) 1 Sam. 16:1: מִלֵּא קַרְנְךָ שֶׁמֶן וְלֵךְ אֲשַׁלְחֶךָ אֶל-יִשָּׁי בֵּית-הַלְחָמִי כִּי־רָאִיתִי בְּבָנָיו לִי מֶלֶךְ
mallē' qarn=akā šemen wə=lēk 'ešālāh=ākā
 fill\IMP[M.SG] horn=yours oil and=go\IMP[M.SG] 1SG-send_out\IPFV=you.OBJ
 'el yišay bēt hallahmī kī rā'i-tī bə=bān=āyw l=ī melek
 to Jesse the_Bethlehemite COMP see\PFV-1SG in=son=PL.his for=me king
 'Fill your horn with oil and go, I will send you to Jesse the Bethlehemite, for I have
 seen a king for me amongst his sons.'

Things become more interesting in cases where something is imposed on the Common Ground. In the following example, David has just sneaked into Saul's camp while Abner was on guard. David then says to Abner that he could have killed Saul. Using *kī*, David suggests that Abner must know that an enemy soldier (David himself) came into the camp, thus emphasizing that Abner did not do a very good job protecting Saul.

- (25) 1 Sam. 26:15: וְלָמָּה לֹא שָׁמַרְתָּ אֶל-אֲדֹנֶיךָ הַמֶּלֶךְ כִּי־בָא אַחֲד הָעָם לְהַשְׁחִית אֶת-הַמֶּלֶךְ אֲדֹנֶיךָ
wə=lāmmā lō' šāmar-tā 'el 'ādōne=kā ham=melek kī
 and=why not guard\PFV-2M.SG to lord=yours the=king COMP
bā' aḥad hā='ām lə=hašhūt 'et ham=melek
 come\PFV[3M.SG] one.of the=army to=destroy\INF OBJ the=king
'ādōne=kā
 lord=yours
 'And why haven't you protected your lord the king, *given that* a soldier came to kill
 the king your lord?'

In each of the examples in (21–25), the Speaker positions a sentential complement close to the Addressee using [+distal] *kī*. Depending on the context, this can have several effects: it may signal to the Addressee that discourse-old information is being referenced (21–22), or the Addressee may be prompted to accommodate some new information in the Common Ground (23–24), or the Speaker may present information as something the Addressee should have known, knowing full well that they do not (25).

As further evidence for the fact that causal *kī* marks Common Ground, it is interesting to see that when a cause consists of partially new information, the new information can be introduced by *wə-hinnē* 'and behold'.³¹ In (26), the theory correctly predicts that *wə-hinnē* should be used instead of mere *ʔ wə* 'and', because it is needed to separate discourse-old from discourse-new information.

³¹ McCarthy (1980: 333–334) claims that *wə-hinnē* 'and behold' can have a causal sense and practically replace *kī*, but the examples are not convincing so I will not compare the two.

- (26) Jdg. 18:9: קוּמָה וְנַעֲלָה עֲלֵיהֶם כִּי רָאִינוּ אֶת־הָאָרֶץ וְהִנֵּה טוֹבָה מְאֹד
qūm-ā wə=n-a'āle 'ālē=hem kī rā'ī-nū 'et
 stand_up\IMP-M.SG and=1PL-go_up\IPFV to=them COMP see\PFV-1PL OBJ
hā=ʾāreṣ wə=hinnē ṭōb-ā mə'ōd
 the=land(F) and=behold good-F.SG very
 '(And the Danites returned to their brothers in Zorah and Eshtaol, and their brothers said to them: "How did it go?" And they said:) "Come on, let's go up against them, *for* we saw their land, *and look*: it's very good!"'

4.1 Backgrounded causal clauses

There are, however, cases of causal *kī* where Common Ground is not being referred to or even imposed. Here the original [+distal] feature of *kī* is often still relevant. There are plenty of cases where the cause is backgrounded, and in that sense placed at a distance from the main topic of conversation. Quite often the *kī*-clause provides the reason for a positive or negative command (16 out of 39 cases without reference to Common Ground), as in the following examples:³²

- (27) Jdg. 13:5b: וּמִזְרָה לֹא־יַעֲלֶה עַל־רֹאשׁוֹ כִּי־יִנְזֹר אֱלֹהִים יְהִיָּה הַנָּעַר מִן־הַבֶּטֶן
ū=mōrā lō' y-a'āle 'al rōš=ō kī nəzīr 'ēlōhūm
 and=razor(M) not 3M-go_up\IPFV[SG] on head=his COMP dedicated.of God
y-ihye han=na'ar min hab=bāṭen
 3M-be\IPFV[SG] the=boy from the=womb
 '(Look, you are pregnant and will give birth to a son,) but a razor shall not come on his head, *for* the boy will be dedicated to God from the womb.'
- (28) Gen. 43:16: הֲבֵא אֶת־הָאֲנָשִׁים הַבֵּיתָה וּטְבַח טֶבֶח וְהָיָן כִּי אֶתִּי יֹאכְלוּ הָאֲנָשִׁים בְּצֹהֲרַיִם:
hābē' 'et hā=ʾānāš-īm hab=bāyət-ā ū=təbōah
 bring\IMP[M.SG] OBJ the=man-PL the=house-ALL and=slaughter\IMP[M.SG]
ṭəbāḥ wə=hākēn kī 'itt=ī y-ō'kəl-ū hā=ʾānāš-īm
 animal and=prepare\IMP[M.SG] COMP with=me 3M-eat\IPFV-PL the=man-PL
b=aš=šāhārāyim
 at=the=noon
 'Bring the men to the house and slaughter an animal and prepare it, *for* the men will eat with me at noon.'

In such cases, the *kī*-clause provides an explanation, but it is most important to the Speaker that the command is followed. For this reason, the explanation in the causal clause can be backgrounded and hence marked with [+distal] *kī*. In the remaining cases,

³²In the data set these are marked as "command". Positive commands: Gen. 21:12; 31:12; 40:15a; 43:16; 1 Sam. 14:39a; 16:11; 23:27; Ruth 3:18a. Negative commands: Gen. 2:17; 21:17; 26:24; 35:17; Jdg. 13:5b; 1 Sam. 4:20; 16:7a; 2 Sam. 13:32a.

the *kī*-clause often provides information that is not crucial for the main story line and can therefore be seen as backgrounded as well.³³

4.2 Bleached causal meaning

Ten causal uses remain unexplained: they introduce clauses whose content is not in the Common Ground, easily accommodated, imposed, or backgrounded.³⁴ In (29), for example, it is not known to the Addressee (the reader) that Rebekah is barren, nor is there any reason why it should be easily accommodated. However, the information is not backgrounded either, since it is picked up at the end of the verse: ‘and Rebekah his wife conceived’.

- (29) Gen. 25:21: וַיַּעֲתֵר יִצְחָק לַיהוָה לְנָכַח אִשְׁתּוֹ כִּי עֲקָרָה הִוא
 way-y-ε‘tar yishāq l=yhwh lənōkah ‘ist=ō kī
 CONJ-3M-pray\PRET[SG] Isaac to=Yahweh on_behalf_of wife=his COMP
 ‘āqār-ā hī
 barren-F.SG she
 ‘And Isaac prayed to Yahweh on behalf of his wife, *since* she was barren(, and God heard his prayer and Rebekah his wife conceived.)’

These ten examples thus require an explanation that is not based on a reinterpretation of the [+distal] feature. One possibility is that the [+distal] feature has bleached away in some causal uses, so that *kī* became usable as a general causal connective. There is a clear developmental path for such bleaching to have occurred. In an earlier stage of the language, there were only causal instances that are derived from a [+distal] feature (such as examples [21–28] discussed here). Speakers then reanalyzed *kī* as a general marker of causation. This enabled the spread to cases where the [+distal] feature does not seem to be interpretable any more, such as (29).

Note that the causal use is the most frequent by far, and is therefore most likely to show such semantic bleaching, on the basis of causal uses that do refer to Common Ground. In the data summary in section 2 it can be seen that exceptions in circumstantial uses of *kī* are limited to causal, causal-adversative, and adversative interpretations. It is noteworthy that there are no exceptions whatsoever in conditional, temporal, resultative, and concessive uses, while they are together twice as frequent than causal-adversative and adversative, which do have exceptions. The lack of exceptions in conditional, temporal, resultative, and concessive uses can therefore not be explained by their lower overall frequency. Instead, the distribution suggests that exceptions first arose due to semantic

³³In the data set these are marked as “backgrounded”: Gen. 5:24; 10:25; 15:16; 21:16; 42:4; Jdg. 4:3; 16:17; 1 Sam. 20:26a; 30:12; 2 Sam. 13:2; 14:15; Ruth 1:6a; 3:17. Alternatively, these examples can be assumed to involve a bleached causal meaning, as discussed in the following subsection.

³⁴In the data set these are marked as “bleached”, for reasons explained below: Gen. 25:21; 37:17; Jdg. 6:30ab; 1 Sam. 4:13; 6:19a; 13:19; 30:6a; 2 Sam. 6:6; 19:27a.

bleaching in the most frequent causal use, which enabled the spread to the closely related causal-adversative and adversative uses through semantic shift.

A critic might argue that if we need to assume semantic bleaching to have occurred for some cases anyway, it is simpler to describe the causal meaning of *kī* as bleached in all instances, including when the [+distal] feature is interpretable. On such a view, the fact that the *kī*-clauses in (21–28) express information that is already in the Common Ground, or is easily accommodated/imposed, is accidental. This argument cannot be maintained. First of all, this alternative is not, in fact, simpler, since reinterpretation of a [+distal] feature is independently needed to explain other circumstantial interpretations in sections 5 and 6 below, and there is no reason why a causal interpretation should not be pragmatically inferred as well. Furthermore, this alternative approach cannot account for the distribution of *kī*. Only 2% of the total number of causal *kī*-clauses is exceptional like (29). A quick look at a dictionary (e.g., Clines 1993–2011; Köhler et al. 2001) shows that this is a much smaller portion of the data than is the case with other causal conjunctions (e.g. *yā'an*). Similarly it can be shown that *kī* is much more frequent with commands than other causal conjunctions. These distributional facts can only be explained if *kī* does not simply have a bleached causal meaning, but if many of the causal instances of *kī* are in fact pragmatically inferred from the general meaning of referring to Common Ground.

5 Adversative and causal-adversative *kī*

An adversative clause provides a contrast. The following example refers back to 1 Sam. 8:19, where the people asked for the second time for a king, after Samuel advised against their first request (1 Sam. 8:5, 10–18). The request is discourse-new in 1 Sam. 8:5, and hence made without *kī* there. In (30), Samuel quotes the second request, and *kī* is used; it thus marks the Common Ground status of the request when it is made for the second time.³⁵ Since the people already asked for a king before, this request is in the Common Ground and here marked by *kī*:

- (30) 1 Sam. 10:19: וַתֹּאמְרוּ לוֹ כִּי-מֶלֶךְ תִּשְׁמָרְנוּ עָלֵינוּ
 wat-t-ō'mar-ū lō kī melek t-āšim 'ālē=nū
 CONJ-2M-say\PRET-PL no³⁷ COMP king 2M-appoint\IPFV[SG] over=us
 'But you said: "no; *but* you shall appoint a king over us!"'

I classified cases where the contrasting clause provides the reason or cause for a preceding negative statement as causal-adversative. Here again, the circumstantial clause is in the Common Ground, because its propositional content has already been introduced in the preceding verse:

³⁵In 1 Sam. 8:19 the second request is made with *kī'im*, a grammaticalized construction which I exclude in this study (see footnote 14). Here and in 1 Sam. 12:12b (3e), the second request is paraphrased and plain *kī* is used. Ruth 1:10 is similar: 'But we will return with you to your people!'

³⁷Reading *lō' lō'* 'no' for *lō' lō'* 'to him' on the basis of other manuscripts and ancient translations; see Elliger & Rudolph (1997).

- (31) Gen. 45:8: לֹא־אַתָּם שְׁלַחְתֶּם אֵתִי הִנֵּה כִּי הֵאֵלָהֶם
 lō' 'attem šlah-tem 'ōt=ī hēnnā kī hā=ēlōhūm
 not you send\PFV-2M.PL OBJ=me here COMP the=god
 'It is not you who have sent me here, *but/because* [it is] the deity.'

A comparison with other adversative strategies again shows that the degree of reference to Common Ground is especially high when *kī* is used. When the adversative clause presents new information, we often find the more neutral conjunction *ʔ wə* 'and, but, ...' instead:³⁸

- (32) Gen. 2:16–17: מִכָּל עֵץ־הַגָּן אָכַל תֹּאכַל: וּמִעֵץ הַדְּעִית טוֹב וְרָע לֹא תֹאכַל מִמֶּנּוּ
 mik=kōl 'eš hag=gān 'ākōl t-ō'kēl ū=mē='eš
 from=all.of tree.of the=garden eat\INFABS 2-eat\IPFV[M.SG] and=from=tree.of
 had=da'at tōb wā=rā' lō' t-ō'kal mimmen=nū
 the=knowledge.of good and=evil not 2-eat\IPFV[M.SG] from=it
 'You may eat from all the trees of the garden, *but* from the tree of the knowledge of good and evil you may not eat.'

A more marked way to introduce an adversative clause with new information content is to use *raq* 'however, but'.³⁹ In the following example, the *raq*-clause provides information that is not yet in the Common Ground:

- (33) Exod. 8:24: אֲנֹכִי אֲשַׁלַּח אֶתְכֶם ... רַק הִרְחַק לֹא־תִרְחִיקוּ לְלֶכֶת
 'ānōkī 'āšallah 'et=kem ... raq harhēq lō'
 I 1SG-send_away\IPFV OBJ=you ... however be_far\CAUS.INFABS not
 t-arhīq-ū lā=lekēt
 2-be_far\CAUS.IPFV-M.PL to=go\INF
 'I will let you go ..., *however*, be sure not to go very far.'

Though most instances of adversative and causal-adversative *kī* make reference to the Common Ground, both categories are characterized by a relatively high number of instances that cannot be explained using a [+distal] feature. In both adversative (34) and causal-adversative (35), the *kī*-clause provides information that is not easily accommodated and cannot be seen as backgrounded either:

³⁸Simply searching for "but" in an English translation yields many more examples where *wə* introduces new information (e.g. Gen. 6:8, 18; 8:9).

³⁹However, *ʔak* 'indeed, just, only, but' regularly refers to discourse-old information content when used adversatively (see Levinsohn 2011: 92–94 for contexts where adversative *ʔak* is used to emphasize a previously introduced point). Though *ʔak* has cognates in Tigre and Ge'ez, I am not aware of a widely accepted etymology. It is tempting to associate this particle, like *kī*, with **ka*, but this is not necessary: the existence of some other particles sensitive to the contents of the Common Ground is not in itself problematic for my hypothesis.

- (34) Jdg. 1:19a: וַיִּרֶשׁ אֶת־הַהָר כִּי לֹא לְהוֹרִישׁ אֶת־יֹשְׁבֵי הָעֵמֶק
 way-y-ōreš 'et hā=hār kī lō' lə=hōrīš 'et
 CONJ-3M-conquer\PRET[SG] OBJ the=hill COMP not to=conquer\INF OBJ
 yōšəb-ē hā=ēmeq
 inhabit\PTCP-M.PL.of the=plain
 'And [Judah] conquered the hill country, *but* they could not conquer the people
 living in the plains ...'
- (35) Gen. 24:3–4: לֹא־תִקַּח אִשָּׁה לְבָנִי מִבְּנוֹת הַכְּנַעֲנִי אֲשֶׁר אֲנִי יֹשֵׁב בְּקִרְבָּן: כִּי אֶל־אֶרֶץ וְאֶל־מִוְלָדֶיהָ
 lō' t-iqqaḥ iššā li=bn=i mib=bən-ōt hak=kəna'ānī
 not 2-take\IPFV[M.SG] woman for=son=mine from=child-F.PL.of the=Canaanite
 'āšer 'ānōkī yōšēb bə=qīrb=ō kī 'el 'arš=i
 REL I live\PTCP[M.SG] in=midst=its COMP to land=mine
 wə='el mōlaqt=i t-ēlēk wə-lāqaḥ-tā iššā
 and=to motherland=mine 2-go\IPFV[M.SG] CONJ-take\MOD-2M.SG woman
 li=bn=i
 for=son=mine
 'You must not acquire a wife for my son from the women of the Canaanites among
 whom I am living, *but/because* you must (instead) go to my country and my
 motherland to take a wife for my son'

These cases are therefore exceptional in the same way as causal (29) above. As with causal *kī*, a diachronic account remains necessary for these cases. Given the relatively low number of occurrences of both adversative and causal-adversative *kī*, it is unlikely that these functions developed directly from the general meaning of *kī*, through a process of semantic bleaching similar to that proposed for causal uses in section 4.2. It is more likely that they are the result of a semantic shift, starting out with the already bleached causal meaning. It is easy to see how a causal meaning could expand to causal-adversative, and eventually lose the necessarily causal interpretation to become plain adversative; such a development is also attested in unrelated languages (cf. Kuteva et al. 2019: 84 and references therein; for discussion of this development in Biblical Hebrew, see Locatell 2017: 247–248 and references therein). In my corpus, bleached meanings are needed for 10 exceptional cases of causal *kī*, 2 cases of causal-adversative *kī*, and 5 cases of adversative *kī*.⁴⁰

⁴⁰These are marked as “bleached” in the data set. For the causal cases, see section 4.2. Causal-adversative cases are Gen. 17:15 and 24:4; adversative cases are Gen. 40:14; Jdg. 1:19a; 4:9a; 1 Sam. 15:35a; 2 Sam. 17:11.

In that sense, these conditional uses are much like the temporal uses of *kī*, and unlike the more general conditional use of אם *im* ‘if’. This particle is used more frequently when it is truly uncertain whether a condition will be met. Thus, *kī* and *im* behave similarly to English *when* and *if*, respectively. This can be seen in the following example with *im*, where it is indeed uncertain for Speaker (and Addressee) whether the condition will be met:

- (37) Gen. 18:26: אם־אמצא בסדם חמשים צדיקים בתוך העיר ונשאתי לכל־המקום בעבורם:
im ^{’-emṣā’} *bi=sdōm* *ḥāmišš-im* *ṣaddiq-im* *bə=tōk* *hā=’ir*
 if 1SG-find\IPFV in=Sodom five-PL righteous-PL in=middle.of the=city
wə-nāšā’-tī *lə=kāl* *ham=māqōm* *ba=’ābūr=ām*
 CONJ-forgive\MOD-1SG for=whole.of the=place for=sake=theirs
 ‘If I find at Sodom fifty righteous in the city, I will forgive the whole place for their sake.’

This can also be seen in the fact that when *kī* and *im* are combined, *kī* introduces a general condition, while *im* introduces a more specific condition (Van der Merwe, Naudé & Kroeze 2017: §40.11(1a)). In such a construction, the condition in the *kī*-clause is more likely to be met than the condition in the *im*-clause:

- (38) Lev. 1:2–3: אדם כִּי־יקריב מכם קרבן ליהוה מן־הבהמה מן־הבקר ומן־העוף תקריבו את־קרבנכם:
 אדם כִּי־יקריב מכם קרבן ליהוה מן־הבהמה מן־הבקר ומן־העוף תקריבו את־קרבנכם:
ādām kī *y-aqrīb* *mikk=em* *qārbān* *l=yhwh* *min*
 man COMP 3M-present\IPFV[SG] from=you offering to=Yahweh from
hab=bahēmā min *hab=bāqār ū=min* *haṣ=šō’n* *t-aqrīb-ū* *’et*
 the=animals from the=herd or=from the=flock 2-present\IPFV-M.PL OBJ
qārbān=kem im *’olā* *qārbān=ō* *min* *hab=bāqār* *zākār*
 offering=yours if burnt_offering offering=his from the=herd male
tāmīm y-aqrīb=ennū
 perfect 3M-present\IPFV[SG]=it.OBJ
 ‘If/When (*kī*) a man amongst you brings a sacrifice to Yahweh, you must bring your offer from the animals of the herd or the flock. If (*im*) it is a burnt offering from the herd, he shall offer a male without blemish.’

This fits with the general meaning described for *kī* above: the circumstantial clause does not introduce entirely unexpected information, but only information which is already expected, or at least easily accommodated given the Common Ground. In (36) above, with *kī*, Jacob’s servant is going towards Esau and will therefore surely meet him, while in (37), with *im*, it is not certain at all that there will be fifty righteous at Sodom. In (38), not all sacrifices are burnt offerings, so the condition in the *im*-clause is met in a subset of the worlds in which the condition in the *kī*-clause is met (therefore *im*, unlike *kī*, can here not be translated as ‘when’).

Temporal clauses with future reference time can be seen as an extreme case of such “expected conditionals”: they are, in a way, conditionals of which the condition is certain to be met at some future point in time:

- (39) Gen. 31:49: יֵצֵף יְהוָה בֵּינִי וּבֵינְךָ כִּי נִסְתָּר אִישׁ מֵרֵעֵהוּ
y-iṣēp yhw h bēn=ī ū=bēn=ekā kī
 3M-guard\JUSS[SG] Yahweh between=me and=between=you COMP
n-issāṭēr ṭš mē=rē=ēhū
 1PL-hide\MID.IPFV one from=companion=his
 ‘May Yahweh watch between me and you *when* we are hidden from each other.’

Temporal clauses with past reference time are similar. They require the described event to have occurred; *kī* cannot be used, for example, for counterfactuals, which are typically marked by לוֹ *lū* ‘if only’.⁴² As the examples below show, temporal *kī*-clauses are often preceded by *wayḥi* ‘and it was’ (40–42,44), but this is not always the case (43).⁴³ In several cases, a *kī*-clause with past reference time is frequentative (i.e., it refers to multiple moments in time; cf. English ‘whenever’), as in (40). This is only contextually inferred, however, and not contributed by *kī*, as this interpretation is not always available (41).⁴⁴

- (40) Jdg. 1:28: וַיְהִי כִּי-יָחֶזֶק יִשְׂרָאֵל וַיִּשֶׂם אֶת-הַכְּנַעֲנִי לְמָס וְהוֹרִישׁ לֹא הוֹרִישׁוּ
wa-y-ḥi kī ḥāzaq yiśrā’ēl
 CONJ-3M-be\PRET[SG] COMP be_strong\PFV[3M.SG] Israel
way-y-āśēm ’et hak=kāna’ānī lā=mas
 CONJ-3M-put\PRET[SG] OBJ the=Canaanite to=forced_labor
wə=hōrēš lō’ hōrīš=ō
 but=conquer\INFABS not conquer\PFV[3M.SG]=it.OBJ
 ‘And *whenever* Israel was strong they would put the Canaanites to work, but they did not totally conquer them.’

⁴²Note that a counterfactual, like a temporal or conditional clause, can be described as augmenting the Common Ground. The only difference is that a counterfactual augments the Common Ground with information known to be false. If *kī* were simply described as *augmenting* the Common Ground, there would be no way to exclude a counterfactual interpretation. But since *kī* is [+distal] and therefore marks information that is in the Common Ground or at least easily accommodated, a counterfactual interpretation is excluded.

⁴³For arguments against analyzing *kī* in *wayḥi kī* as a complementizer/nominalizer (i.e., ‘and it was *that*’ instead of ‘and it was *when*’), see Locatell (2017: 236–237). There does not appear to be a difference in the contribution of *kī* when preceded by וַיְהִי *wə-hāyā* ‘and it will be’ or וַיִּהְיֶה *wa-yḥi* ‘and it was’, compared to when *kī* stands alone. The contribution of these temporal markers can be seen as shifting the reference time (e.g. Van der Merwe, Naudé & Kroeze 2017: §40.24–25) independent from the discursive contribution made by *kī*.

⁴⁴A frequentative interpretation is also possible in Jdg. 2:18a; 12:5; 16:16; 2 Sam. 6:13; and we have a durative interpretation (referring to a longer stretch of time; cf. English ‘while’) in 1 Sam. 1:12; 17:48. Such interpretations are excluded in Gen. 6:1; 26:8; 31:37; 43:21; 44:24; Jdg. 6:5, 7; 8:1; 1 Sam. 14:29b; 22:22a; 2 Sam. 4:10, 11; 7:1, 12; 19:26.

- (41) Gen. 27:1: וַיְהִי כִּי־זָקֵן יִצְחָק ... וַיִּקְרָא אֶת־עֵשָׂו בְּנֹו הַגָּדֹל
wa-y-ahī kī zāqēn yiṣḥāq ... way-y-iqrā
 CONJ-3M-be\PRET[SG] COMP be_old\PFV[3M.SG] Isaac ... CONJ-3M-call\PRET[SG]
 'et 'ešāw bən=ō hag=gādōl
 OBJ Esau son=his the=big
 'And *when* Isaac was old ..., he called Esau, his oldest son.'

To return to the second way in which temporal and conditional *kī* interacts with Common Ground: in most instances of temporal or conditional *kī*, the event or condition described in the circumstantial clause is already salient in the surrounding discourse, and in that sense tracked by both Speaker and Addressee. Consider:

- (42) Gen. 44:24: וַיְהִי כִּי עָלִינוּ אֶל־עַבְדְּךָ אָבִי
wa-y-ahī kī 'ālī-nū 'el 'abd=akā 'āb=i
 CONJ-3M-be\PRET[SG] COMP go_up\PFV-1PL to servant=yours father=mine
 'And *when* we went up to your servant my father, ...'
- (43) Jdg. 2:16–18: וַיִּקֶּם יְהוָה שְׁפָטִים ... וְכִי־יִהְיוּ יְהוָה לְהֵם שְׁפָטִים
way-y-āqem yhw h šōpāt-īm ... wə=kī
 CONJ-3M-establish\PRET[SG] Yahweh judge-PL ... and=COMP
hēqīm yhw lā=hēm šōpāt-īm
 establish\PFV[3M.SG] Yahweh for=them judge-PL
 'And Yahweh established judges ... *Whenever* Yahweh established judges for them, (Yahweh would be with the judge ...)'
- (44) 1 Sam. 14:27–29b: וַיָּשָׁב יָדֹו אֶל־פִּיו וַתֹּאמֶרְנָה⁴⁵ עֵינָיו: ... אָרו עֵינַי כִּי טַעַמְתִּי מֶעֶט דְּבַשׁ הַזֶּה:
way-y-āšeb yād=ō 'el pī=w
 CONJ-3M-return\CAUS.PRET[SG] hand=his to mouth=his
wat-t-ā'ōr-anā 'en=āyw ... 'ōr-ū 'en-ay
 CONJ-F-light_up\PRET-3PL eye(F)=DU.his ... light_up\PFV-3PL eye-DU.mine
kī ṭā'am-tī mā'aṭ dābaš haz=ze
 COMP taste\PFV-1SG little.of honey the=this
 'And [Jonathan] returned his hand to his mouth and his eyes lit up. (So a soldier spoke up and said, "Your father swore the army, saying, 'Cursed is the man who eats anything today'" (...) But Jonathan said: "My father has brought misfortune on the land. See that) my eyes lit up *when* I tasted a little of this honey!'

⁴⁵Consonantal text: וַתֹּאמֶרְנָה *wat-t-ir'e-nā* CONJ-3F-see\PRET-PL 'and [his eyes] saw'. I follow the reading of the vocalized text, which was standardized in the Middle Ages and is here corroborated by various manuscripts and ancient translations (Elliger & Rudolph 1997).

In (42), it is clear from the context to Joseph (the Addressee) that his brothers (the Speaker) have been to their father. It is therefore assumed to be part of the Common Ground, and *kī* can be used to refer to this event. In (43) this is even clearer, as the fact that Yahweh established judges has been introduced just a few sentences before. Example (44) is similar: the event described by the *kī*-clause is introduced in v. 27, and v. 28 (only given in translation) shows that the Addressee (the soldier) is aware of it as well.

To conclude this subsection: there are two ways in which temporal and conditional *kī*-clauses interact with the Common Ground. As we just saw, in many cases the event or condition is already tracked by both the Speaker and the Addressee, independent of whether it is likely to occur or have occurred. More importantly, however, in all instances the event or condition is certain or likely to occur or have occurred. As a result, if the event or condition is not yet in the Common Ground, it is at least easily accommodated. In this way these cases support the hypothesis about the general discursive contribution of *kī*.

6.2 Resultative clauses

Resultative clauses (sometimes called “consequential clauses”) describe the result of the event described in the matrix clause; in English, resultative *kī* can often be translated with (so) *that*. When the result is also the purpose for a volitional act described in the matrix clause, a resultative clause is quite similar to a causal clause:

- (45) Gen. 31:36: מָה חָטָאתִי כִּי דָלַקְתָּ אַחֵרַי: *mah haṭṭā'-tī kī dālaq-tā 'ahār=āy*
 what sin\PFV-1SG COMP chase\PFV-2M.SG behind=me
 ‘How have I (Jacob) sinned *that* you (Laban) have chased after me?’

In (45), it is in the Common Ground that the Addressee chased after the Speaker, as the Addressee has just caught up with the Speaker when (45) is uttered. Note that there is no reason why a purpose or resultative clause in general should refer to a result in the Common Ground, as the following example with another resultative connective, לָמַעַן *ləma'an*, shows:

- (46) Gen. 37:22: וַיֹּאמֶר אֲלֵהֶם | רְאוּבֵן אֶל־תִּשְׁפְּכוּ־דָם הַשְּׁלִיכוּ אֹתוֹ אֶל־הַבּוֹר הַזֶּה אֲשֶׁר בְּמִדְבָּר וַיֵּד: *way-y-ō'mer 'ālē=hem rə'ūbēn 'al t-išpək-ū dām*
 CONJ-3M-say\PRET[SG] to=them Reuben not 2-shed\JUSS-M.PL blood
 וַיִּשְׁלֹחַ וַיִּחְוֶיבֵּן לָמַעַן הַצִּיל אֹתוֹ מִיָּדָם לְהָשִׁיבוּ אֶל־אָבִיו: *hašlik-ū 'ōt=ō 'el hab=bōr haz=ze 'āšer b=am=miḏbār wə=yāḏ*
 throw\IMP-M.PL OBJ=him to the=pit the=this REL in=the=desert but=hand
 וַיִּשְׁלַח וַיִּשְׁלַח וַיִּשְׁלַח וַיִּשְׁלַח וַיִּשְׁלַח וַיִּשְׁלַח וַיִּשְׁלַח וַיִּשְׁלַח וַיִּשְׁלַח וַיִּשְׁלַח *'al t-išlah-ū b=ō ləma'an haššil 'ōt=ō*
 not 2-send\JUSS-M.PL against=him in_order_to rescue\INF OBJ=him
 מִיָּדָם לְהָשִׁיבוּ אֶל־אָבִיו: *miy=yāḏ=ām la=haššib=ō 'el 'ābī=w*
 from=hand=theirs to=return\CAUS-INF=him to father=his

‘But Reuben said to them, “Don’t shed blood; throw him into this pit in the desert but do not stretch out your hand against them,” *in order to* rescue him out of their hand to return him to his father.’

In (46), the fact that Reuben tries to save Joseph is not yet known to the Addressee (the reader). It can be accommodated based on the contents of the direct speech report, but even this is not always the case. For instance, in (47) with the purposive construction *lā* ‘to’ + infinitive, there is no reason in particular to think that Laban should go shear his sheep at this moment:

- (47) Gen. 31:19: וּלְבֹן הַלֵּךְ לִגְזֹז אֶת־צֹאֲנוֹ
 wə=lābān hālāk li=gzōz ’et šō’n=ō
 and=Laban go\PFV[3M.SG] to=shear\INF OBJ flock=his
 ‘And Laban had gone *in order to* shear his sheep.’

Nevertheless, we do not find such cases with resultative *kī*. Even in cases with future reference time, the resultative clause refers to the Common Ground because the prospected result has already been discussed. For instance, in (48), though the resultative is an irrealis with future time reference, it refers directly back to Saul’s proposal in the previous verse.

- (48) 1 Sam. 18:18: מִי אֲנִי ... כִּי־אֶהְיֶה חָתָן לְמֶלֶךְ
 mī ’ānōkī ... kī ’-ehye ḥātān l=am=melek
 who I ... COMP 1SG-be\IPFV son_in_law to=the=king
 ‘(Saul said to David: “Here is my oldest daughter Merab; I want to give her to you in marriage ...” But David said to Saul:) “Who am I ... *that* [you say that] I should be the king’s son-in-law?”’

This supports the hypothesis that *kī* is marked for reference to the Common Ground.

6.3 Concessive clauses

With a concessive clause (English *though*) the Speaker concedes some information to the Addressee, but at the same time denies that this information is incompatible with the assertion made in the matrix clause:

- (49) 2 Sam. 12:12: כִּי אָתָּה עָשִׂיתָ בְּסֵתֶר וְאֲנִי אֶעֱשֶׂה אֶת־הַדָּבָר הַזֶּה נֶגְדִּי כָל־יִשְׂרָאֵל
 kī ’attā ’āšī-tā b=as=sāter wa=’ānī ’-ešše ’et had=dābār
 COMP you do\PFV-2M.SG in=the=secret and=I 1SG-do\IPFV OBJ the=thing
 haz=ze neḡeḏ kāl yiśrā’ēl
 the=this before all.of Israel
 ‘*Though* you have acted in secret, I will do this before all of Israel.’

A concessive usually presupposes that the Addressee knows or can easily accommodate the conceded information. For instance, in English I can utter (50) only to someone of whom I know that they share my belief about France's likelihood to win; I have to assume that *France won't win this World Championship* is in the Common Ground or easily accommodated.

(50) *Though France won't win this World Championship, they are a treat to watch.*

The fact that concessive clauses always refer to Common Ground makes it pointless to compare *kī* with other markers of concessive clauses, such as כִּי *im* (Job 9:15; Jer. 15:1) or וְ *wə* (e.g. Jdg. 16:15). However, the fact that the cases with *kī* all refer to information in the Common Ground or easily accommodated is of course in line with my proposal.

6.4 Low-frequency circumstantial clauses: summary

This section has reviewed four types of less frequent circumstantial *kī*-clauses: temporal, conditional, resultative, and concessive clauses. It is particularly important that these lower frequency uses of *kī* adhere to the predicted patterns, since they are a priori less likely to have bleached and lost the [+distal] feature. The data reveal that in these *kī*-clauses the information provided in the clause is indeed always in the Common Ground, easily accommodated, or (rarely) imposed on the Common Ground for a discursive effect. This confirms the hypothesis that *kī* still has a [+distal] feature, which is interpreted as referring to the Addressee, and thus to the Common Ground.

7 Standalone *kī*-clauses

As is well-known and seen in the previous sections, *kī* usually connects two clauses. After classifying all instances of *kī* for the relation it establishes between the two clauses, some instances remained for which it is not clear that *kī* really connects two clauses. Though these cases have often been seen as deviant in the literature on Biblical Hebrew (for instance, they prompt [Park 2016](#) to propose that *kī* is a “nominalizer”), they are actually to be expected when we look at similar clausal connectives in other languages. For instance, English *that* can be used in matrix clauses (see [1] above), as can Romance complementizers derived from Latin *quod* ([Cruschina & Remberger 2017](#); [Kocher 2022](#)); for more discussion, see [Mithun \(2016\)](#). Scholars of Biblical Hebrew have described similar instances with *kī* as “emphatic” or “asseverative” (e.g. [Joüon & Muraoka 2006](#): §164; [Miller 2003](#): 103–116; [Holmstedt 2010](#): 76, 85, 92). Because these terms are met with criticism for being vague (e.g. [Bandstra 1982](#); [Aejmelaeus 1986](#); [Locatell 2017](#)), I have classified these instances as “standalone” to avoid committing to any analysis upfront. A closer look reveals that there are three ways in which standalone *kī* can be used: to introduce oaths (section 7.1), conducive and rhetorical polar questions (section 7.2), or exclamatives (section 7.3). Each of these functions can reasonably be called “emphatic”, but it is valuable to be precise what

kinds of emphasis can be provided by *kī*, exactly, to prevent this category from becoming a universal catch-all.

7.1 Oaths

Oaths can be described as sincere and earnest speech acts meant to assure the Addressee of a certain assertion or promise (Conklin 2011: 2). They are typically accompanied by what Conklin (2011: 13–30) calls an “authenticating element” that is meant to assure the Addressee of the Speaker’s sincerity, such as ‘life of Yahweh’ in (51).⁴⁶

- (51) 2 Sam. 12:5: חַי־יְהוָה כִּי בֶן־מָוֶת הָאִישׁ הָעֹשֶׂה זֹאת: *hay yhw h kī ben māwet hā=’iš hā=’ōsé zōt*
 life.of Yahweh COMP son.of death the=man the=do\PTCP[M.SG] this
 ‘By the life of Yahweh, *that* the man who does this is a dead man!’

There are 17 cases of standalone *kī* introducing an oath in my corpus, of which only six refer to discourse-old information content.⁴⁷ For example, in (52), the previous verse has already made the question of when the people will stop pursuing their brothers a topic of discussion, and therefore tracked in the Common Ground:

- (52) 2 Sam. 2:27a: וַיֹּאמֶר יוֹאָב חַי הָאֱלֹהִים כִּי לֹא דִבַּרְתָּ כִּי אֶזְמַח מִן־הַבֹּקֶר נַעֲלֶה הָעָם אִישׁ מֵאַחֲרָי אָחִיו: *way-y-ō’mer yō’āb hay hā=’ēlōhīm kī lūlē’ dibbar-tā*
 CONJ-3M-say\PRET[SG] Joab life.of the=god COMP had_not speak\PFV-2M.SG
kī ’āz mē=hāb=bōqer na’ālā hā=’ām ’iš
 COMP then from=the=morning go_up\MID.PFV[3M.SG] the=people(M) one
mē=’ahāre ’āhī=w
 from=behind brother=his

‘(Abner called out to Joab: “... How long won’t you tell the people to return from after their brothers?”) And Joab said: “By the life of the deity, ([I swear] *that*) had you not spoken, (*that*) then [only] from the morning onwards would the people have ceased [pursuing], each from behind his brother.”’

Usually, however, the content of the oath is not discourse-old, and cannot be accommodated either. Instead, it is imposed on the Common Ground.⁴⁸ This is also how I analyze (51) above. It is precisely the imposition that triggers the interpretation as an oath. Forcing the Addressee to accept an assertion in the Common Ground, the Speaker effectively assures the Addressee of their own sincerity and commitment to this assertion, which is precisely what an oath does (Conklin 2011: 2).

⁴⁶ Besides *kī*, a number of other particles can be used to introduce oaths (Conklin 2011: 60–65), but they do not appear often enough to allow for a comparison with *kī*. For this reason I will only discuss oaths with *kī* in this subsection.

⁴⁷ 1 Sam. 20:13; 25:34ab; 29:6a; 2 Sam. 2:27a; Ruth 1:17.

⁴⁸ 1 Sam. 14:39b; 20:3b; 20:12; 26:10, 16; 2 Sam. 3:9ab, 35; 12:5; 15:21ab.

Conklin's explanation for the use of *kī* to mark oaths is reminiscent of Ross's (1970) performative hypothesis: *kī* would be a remnant of an originally longer formula (*I swear that ...*) after elision of the predicate (Conklin 2011: 59). There are well-known problems with this approach in its most basic form (see Speas & Tenny 2003: 338 for discussion), but even if we were to accept it, some questions remain. For example, it is unclear why the complementizer would not have been elided together with the predicate in oath formulas, with only intonation serving to distinguish the exclamative from a declarative (as in some rhetorical questions, like *I said something funny?*, and interrogatives more generally in languages like Italian). The analysis proposed in the present article provides an explanation why the complementizer was retained: it is crucial to oaths that they impose information on the Common Ground, and this aspect is marked by the [+distal] complementizer.

7.2 Conducive and rhetorical questions

There are five instances in my corpus of conducive and rhetorical questions using the polar interrogative marker *hā* followed by *kī*, two of which are negated (*hā-lō' kī*).⁴⁹ In one instance (2 Sam. 9:1), *hā-kī* introduces a regular question which happens to nominalize a clause of which the content is under discussion in the Common Ground;⁵⁰ this instance is excluded here and classified as a subject clause. I will rely on Moshavi (2009) for my analysis of the five instances under discussion here; see there for more background and references on these types of questions.

A rhetorical question has the form of an interrogative but is, at the discursive level, an implicit assertion rather than a request for information (Moshavi 2009: 32). The implicit assertion contributed by a rhetorical polar question is the negation of its propositional content: *Are you the president?* implies that you are not the president (Moshavi 2009: 33). A conducive question is similar to a rhetorical question in that the Speaker has a certain prior belief regarding the correct answer, and may not expect an answer. However, unlike a rhetorical question it does not function as an implicit assertion. For example, *Is that you, Henry?* does not imply that you are Henry but merely conveys an expectation (Moshavi 2009: 38). Moshavi describes the discursive functions of conducive questions in Biblical Hebrew as (a) confirming a belief of the Speaker, (b) expressing surprise, (c) showing the Addressee that the Speaker knows something to be true, and (d) drawing attention to a fact (2009: 38 n. 38).

The questions with *hā-kī* 'is [it] that' are conducive, implying that the information content of the *kī*-clause is true:

⁴⁹Without negation: Gen. 27:36; 29:15; 2 Sam. 23:19. With negation: 1 Sam. 10:1; 2 Sam. 13:28.

⁵⁰Similar to one more example outside my corpus, Job 6:22.

- (53) Gen. 27:36: הֲכִי קָרָא שְׁמוֹ יַעֲקֹב וַיַּעֲקֹבֵנִי זֶה פַּעַמִּים
hă=kī qārā šəm=ō ya'āqōb way-y-a'qəb=ēnī
 Q=COMP call\PFV[3M.SG] name(M)=his Jacob CONJ-3M-deceive\PRET[SG]=me
zε pā'ām-ayim
 this time-DU
 'Isn't his name Jacob?⁵¹ He has deceived me these two times!'

In the case of conducive questions with *kī*, the information content is not only implied but also well-known to be true, and thus in the Common Ground. Compared to conducive questions without *kī*, the questions with *kī* convey a much stronger belief with respect to the expected answer. In (54) without *kī* the Speaker is much less certain that Saul should now be considered a prophet than that the Speaker in (53) is certain about Jacob's name. This explains why *kī* can be used in (53) but not (54): only in (53) can the propositional content be assumed to be in the Common Ground. In (54), the Speaker does not even want to impose it on the Common Ground.

- (54) 1 Sam. 10:11: וַיְהִי כִּלְיוֹדְעוֹ מֵאַתְמוֹל שְׁלֹשׁ וַיְרָא וְהִנֵּה עִם־נְבִיאִים נִבָּא וַיֹּאמֶר הֲעַם אִישׁ אֶל־רֵעֵהוּ:
 מִה־זֶּה הִיָּה לְבִן־קִישׁ הַגִּם שָׂאוֹל בְּנִבְיָאִים:
wa-y-əhī kāl yōdā=ō mē='ittamōl šilšōm
 CONJ-3M-be\PRET[SG] all.of know\PTCP[M.SG]=him from=before
way-y-ir'-ū wə=hinnē 'im nəbī'-im nibbā'
 CONJ-3M-see\PRET-PL and=behold with prophet-PL prophesy\PTCP[M.SG]
way-y-ō'mer hā=ām 'iš 'el rē'=ēhū mah zε
 CONJ-3M-say\PRET[SG] the=people(M) one to companion=his what this
hāyā lə=ben qīš hă=gam šā'ul b=an=nəbī'-im
 be\PFV[3M.SG] to=son.of Kish Q=also Saul in=the=prophet-PL
 'And when all who knew [Saul] from before saw how he prophesied with prophets, the people said to each other: "What happened to the son of Kish? Is Saul also among the prophets?"'

The questions with *hă-lō' kī* 'isn't [it] that' are rhetorical. Due to double negation (once for *lō'* 'not' and once for the rhetorical question), these questions also imply that the information content of the *kī*-clause is true:

⁵¹The name Jacob translates as 'he-will-deceive'. Note that my translation uses a rhetorical question, and thus adds negation. Another option is "No wonder his name is Jacob" (NLT; cf. also NET). One can also take this *kī* as causal ('Is it *because* his name is Jacob that ...'), which is also possible in Gen. 29:15 but not in 2 Sam. 23:19.

- (55) 1 Sam. 10:1: וַיִּקַּח שְׂמוּאֵל אֶת־פֶּךָּ הַשֶּׁמֶן וַיִּצֹק עַל־רֹאשׁוֹ וַיִּשְׁקָהוּ וַיֹּאמֶר הֲלוֹא כִי־מָשַׁחְךָ יְהוָה
 :עַל־נִחְלָתוֹ לְנָגִיד:
way-y-iqqah šamū'el 'et pak haš=šemen way-y-išōq
 CONJ-3M-take\PRET[SG] Samuel OBJ flask.of the=oil CONJ-3M-pour\PRET[SG]
'al rōš=ō way-y-iššāq=ēhū way-y-ō'mer hā=lō'
 on head=his CONJ-3M-kiss\PRET[SG]=him CONJ-3M-say\PRET[SG] Q=not
kī māsāh=ākā yhw h al nahālāt=ō la=nāgīd
 COMP anoint\PFV[3M.SG]=you.OBJ Yahweh over inheritance=his for=leader
 'Then Samuel took a flask of oil and poured it on [Saul]'s head, and he kissed him
 and said: "Has Yahweh *not* anointed you as leader over his inheritance?"

In the other rhetorical question, Common Ground is imposed by making an implicit assertion (2 Sam. 13:28). A rhetorical question does not leave room for the Addressee to reject the implied assertion, and thus has the effect of imposing it on the Common Ground. Though the use of *kī* is not necessary to form these types of questions (see [Moshavi 2009](#) for many examples without *kī*), it is still well-suited because of this interaction with the Common Ground.

7.3 Exclamatives

The third way in which standalone *kī* is used is to form exclamatives. Exclamatives are sentences that "express the speaker's affective response to a situation" ([Michaelis 2001](#): 1039). As with conducive and rhetorical questions, a proposition can be recovered from an exclamative: *What nice weather it is!* implies that it is nice weather. Crucially, the Speaker of an exclamative assumes this propositional content to be in the Common Ground ([Zanutini & Portner 2003](#)): *What nice weather it is!* cannot be used to convince the Addressee that it is nice weather. This distinguishes exclamatives from both declarative sentences, which make no such assumption, and rhetorical questions, which impose information on the Common Ground rather than assuming it is already shared with the Addressee. Nevertheless, this reference to Common Ground makes *kī* a natural element to mark exclamatives (cf. [Mithun 2016](#) and [Staps & Rooryck 2023](#): 10–12 for comparative parallels).

In the examples found in the Hebrew Bible, exclamatives with *kī* express shock/outrage (56)⁵² or commitment (57) with respect to the recoverable proposition, though there is no reason to think other emotions could not be expressed this way as well.

⁵² Some translations take these instances of *kī* as causal, providing a reason for the following clause ('I will go down ...': KJV; ESV), but modern versions provide support for an exclamative reading (NASB/AMP: "the outcry ... is *indeed* great, and their sin is *exceedingly* grave"). For other examples, see 1 Sam. 17:28 and perhaps Gen. 45:26a, if we assume that the brothers are in part speaking to each other (this explanation is not ideal, but I have no better alternative at this point).

- (56) Gen. 18:20ab: זַעֲקַת סְדֹם וְעִמְרָה כִּי־רַבָּה וְחַטָּאתָם כִּי כְבֹדָהּ מֵאֵד:
za'āqa-t sādōm wa=āmōrā kī rābb-ā wə=ḥattā't=ām
 outcry(F)-of Sodom and=Gomorrah COMP be_great\PFV-3F.SG and=sin(F)=theirs
kī kāḇad-ā mə'ōd
 COMP be_heavy\PFV-3F.SG very
 'That the outcry of/concerning Sodom and Gomorrah is so great! And *that* their sin is so heavy!'
- (57) 1 Sam. 10:24: הֲרֵאיתֶם אֲשֶׁר בָּחַר־בּוֹ יְהוָה כִּי אֵין כָּמֹהוּ בְּכָל־הָעָם
har=rə'i-tem 'āšer bāḥar b=ō yhw h kī 'en
 Q=see\PFV-2M.PL REL choose\PFV[3M.SG] on=him Yahweh COMP not_exists
kāmō=hū bə=kāl hā=ām
 like=him in=all.of the=people
 'Do you see whom Yahweh has chosen? *That* no one like him (=Saul) exists among all the people!'

These examples refer to Common Ground in the following way. In (56), God speaks to Abraham. Abraham has already had previous contact with Sodom in Genesis 13–14, where he has been able to see that the Sodomites are *rā'im wə-ḥattā'im* 'wicked and sinners' (Gen. 13:13). This allows the Speaker to presuppose this information, and thus permits the interpretation as an exclamative. Example (57) is spoken by Samuel after he has proclaimed Saul king. Similar to (56), the claim that there is no one like Saul among the people can be accommodated by the Addressee (the people) from the fact that Saul has just been made king.

It is also possible to express commitment towards a wish, which has the effect of strengthening a wish (58).⁵³ It should be noted that in these cases the fact that the Speaker has a certain desire is discourse-old: in (58), it is already clear to the Addressee that the Speaker wants the spoil to be divided equally. It thus does not appear to be possible to strengthen just any wish with *kī*.

- (58) 1 Sam. 30:24: כִּי כְחֶלֶק! הִירָד בַּמִּלְחָמָה וְכְחֶלֶק הַיָּשָׁב עַל־הַבָּלָיִם יִחָדּוּ יְחֻלְקוּ:
kī kə=ḥēleq hay=yōrēd b=am=miḥāmā ū=kə=ḥēleq
 COMP like=part.of the=go_down\PTCP[M.SG] in=the=battle and=like=part.of
hay=yōšēb 'al hak=kēl-īm yaḥdāw y-aḥālōq-ū
 the=sit\PTCP[M.SG] on the=item-PL together 3M-divide\IPFV-PL
 '("Since they didn't go with me, we will not give them from the spoil. ...") But David said: "No! ...) *That* as the part of he who goes down in battle, so be the part of he who remains with the equipment! Together they shall divide it."

⁵³Also 1 Sam. 8:9; 14:44 (if not an oath); 25:28a.

Exclamatives are clearly emphatic, but it is a much narrower category than “emphatic” or “asseverative” *kī*. Before classifying an instance of *kī* as an exclamative it must be shown that the propositional content is in the Common Ground or easily accommodated by the Addressee. This more precise description of the function of *kī* can thus help prevent overly liberal use of the notion of “emphatic” *kī*.

8 Conclusion

The primary function of Biblical Hebrew *kī* is to mark reference to Common Ground, which includes reference to easily accommodated and imposed information content. Deviations from this general pattern mostly occur in the most common function, causal *kī* (section 4). I therefore conclude that some semantic bleaching must have occurred here (section 4.2), which subsequently transferred to the semantically closely related categories of causal-adversative and adversative *kī* (section 5). In all other functions, the vast majority of instances refer to the Common Ground. Though grammaticalization is required to account for the use of *kī* to introduce subject and object clauses (section 3), we do not need to assume intricate grammaticalization paths to account for the various circumstantial uses of *kī* (section 6; contra e.g. Locatell 2017, 2020). These are more economically described as pragmatically inferred uses based on the general function of marking Common Ground; since some of these functions are very infrequent, assuming semantic shifts for which we do not have evidence is problematic. Taking all of this into account, I propose a description of *kī* with only three distinct functions:

1. Referring to Common Ground (including easily accommodated and imposed information content)
 - (a) As a complementizer introducing subject and object clauses
 - (b) When connecting two clauses: introducing circumstantial clauses (adversative, causal, causal-adversative, concessive, conditional, resultative, temporal)⁵⁴
 - (c) When standalone: introducing oaths, conducive and rhetorical questions (with *hā*), and exclamatives
2. Bleached causal meaning (‘because’, ‘for’, etc.)
3. Bleached adversative meaning (‘but’), developed from causal via causal-adversative (‘not X, *but/because* Y’)

Some authors, Aejmelaeus (1986: 193–194) most verbosely, have wondered how native speakers could have distinguished the many uses of *kī*. Though this study is based on texts and not native speaker judgments, the three functions above suggest a straightforward answer. The most common and default function of *kī* is to mark Common Ground. When the Addressee cannot accommodate the information content of a *kī*-clause, this triggers

⁵⁴Given that some of these types are very infrequent, it is conceivable that there are still other types of circumstantial clauses that could be marked by *kī*, which have not made their way into the corpus. This is fine, as long as the information content is in the Common Ground and the discursive function can be inferred from context.

them to use one of the bleached meanings instead, choosing between causal and adversative based on context. In its default function of marking Common Ground, one of the three subfunctions can be selected based on simple cues: the existence of a matrix predicate or the place of the *kī*-clause in subject position for complementizer *kī*, authenticating elements for oaths, the interrogative particle for questions, and intonation for exclamatives; anything else is circumstantial. The appropriate kind of circumstantial meaning can be selected based on context.

The function of marking Common Ground can be explained in the framework of Staps & Rooryck (2023). In their model, information content is conceptualized as located in an abstract space, in which it can be “near” or “far from” Speaker and Addressee. I have argued that the Biblical Hebrew particle *kī* has inherited a [+distal] feature, which on the discourse level positions information content “far” from the Speaker, and therefore “near” the Addressee. The effect is that it marks Common Ground, which consists of information content known to the Addressee. The analysis presented here thus lends further support to the idea that information content is conceptualized as positioned in an abstract space (Staps & Rooryck 2023), and that highly grammaticalized function words like clausal connectives can retain features like [+distal] that receive a new interpretation in this abstract context.

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References

- Aejmelaeus, Anneli. 1986. Function and interpretation of כִּי in Biblical Hebrew. *Journal of Biblical Literature* 105(2). 193–209. <https://doi.org/10.2307/3260389>.
- Axel-Tober, Katrin. 2017. The development of the declarative complementizer in German. *Language* 93(2). e29–e65. <https://doi.org/10.1353/lan.2017.0030>.
- Bandstra, Barry Louis. 1982. *The syntax of particle ky in Biblical Hebrew and Ugaritic*. Yale University dissertation.
- Bar-Asher Siegal, Elitzur A. 2013. *Introduction to the grammar of Jewish Babylonian Aramaic* (Lehrbücher Orientalischer Sprachen 3). Münster: Ugarit-Verlag.
- Benigni, Antonella. 1999. The Biblical Hebrew particle כִּי from a discourse analysis perspective. *Zeitschrift für Althebraistik* 12(2). 126–145.
- Bolinger, Dwight. 1972. *That's that* (Janua Linguarum 155). The Hague: Mouton.
- Budde, Karl. 1889. Zu Habakuk 2, 3 ff. [On Habakkuk 2:3ff.] *Zeitschrift für die Alttestamentliche Wissenschaft* 9(1). <https://doi.org/10.1515/zatw.1889.9.1.155>.

- Claassen, Walter T. 1983. Speaker-oriented functions of *kî* in Biblical Hebrew. *Journal of Northwest Semitic Languages* 11. 29–46.
- Clark, Herbert H. 1996. *Using language*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9780511620539>.
- Clines, David J. A. (ed.). 1993–2011. *The dictionary of Classical Hebrew*. Sheffield: Sheffield Academic Press.
- Colasanti, Valentina & Martina Wiltschko. 2019. Spatial and discourse deixis and the speech act structure of nominals. In *Proceedings of the 2019 annual conference of the Canadian Linguistics Association*, 1–14. <http://hdl.handle.net/2262/100137>.
- Conklin, Blane. 2011. *Oath formulas in Biblical Hebrew* (Linguistic Studies in Ancient West Semitic 5). Winona Lake, IN: Eisenbrauns.
- Cruschina, Silvio & Eva-Maria Remberger. 2017. Before the complementizer: Adverb types and root clause modification. In Martin Hummel & Salvador Valera (eds.), *Adjective adverb interfaces in Romance* (Linguistics Today 242), 81–109. Amsterdam: John Benjamins. <https://doi.org/10.1075/la.242>.
- Dancygier, Barbara & Eve Sweetser. 2000. Constructions with *if*, *since*, and *because*: Causality, epistemic stance, and clause order. In Elizabeth Couper-Kuhlen & Bernd Kortmann (eds.), *Cause, condition, concession, contrast: Cognitive and discourse perspectives* (Topics in English Linguistics 33), 111–142. Berlin: Mouton de Gruyter. <https://doi.org/10.1515/9783110219043-006>.
- de Cuba, Carlos. 2023. Relatively nouny? In Katrin Axel-Tober et al. (eds.), *On the nominal nature of propositional arguments*, 19–42. Hamburg: Buske.
- Dhorme, Édouard. 1967. *A commentary on the book of Job*. Nashville: Thomas Nelson.
- Diessel, Holger. 1999. *Demonstratives. Form, function, and grammaticalization* (Typological Studies in Language 42). Amsterdam: John Benjamins. <https://doi.org/10.1075/tsl.42>.
- Elliger, Karl & Wilhelm Rudolph. 1997. *Biblia Hebraica Stuttgartensia*. 5th edn. Stuttgart: Deutsche Bibelgesellschaft.
- Elsness, Johan. 1984. *That* or zero? A look at the choice of object clause connective in a corpus of American English. *English Studies* 65(6). 519–533. <https://doi.org/10.1080/00138388408598357>.
- Farkas, Donka F. & Kim B. Bruce. 2010. On reacting to assertions and polar questions. *Journal of Semantics* 27(1). 81–118. <https://doi.org/10.1093/jos/ffp010>.
- Follingstad, Carl M. 2001. *Deictic viewpoint in Biblical Hebrew text: A syntagmatic and paradigmatic analysis of the particle כִּי (kî)* (Journal of Translation and Textlinguistics). Dallas, TX: SIL International.
- Goldenberg, Gideon. 1991. On direct speech and the Hebrew Bible. In Karel Jongeling, Heleen Murre-van den Berg & Lucas Van Rompay (eds.), *Studies in Hebrew and Aramaic syntax* (Studies in Semitic Languages and Linguistics 17), 79–96. Leiden: Brill. https://doi.org/10.1163/9789004348332_so10.
- Gross, Walter. 1991. Satzfolge, Satzteilfolge und Satzart als Kriterien der Subkategorisierung hebräischer Konjunktionalsätze, am Beispiel der כִּי-Sätze untersucht.

- [Sentence order, clause order, and sentence type as criteria for the subcategorization of Hebrew conjunctive clauses, studied using the example of ו- clauses]. In Walter Gross, Hubert Irsigler & Theodor Seidl (eds.), *Text, Methode und Grammatik: Wolfgang Richter zum 65. Geburtstag*. [Text, method, and grammar: Wolfgang Richter on his 65th birthday], 97–119. St. Ottilien: EOS Verlag.
- Gzella, Holger. 2008. Zum periphrastischen Infinitiv in Genesis viii 5. [On the periphrastic infinitive in Genesis 8:5]. *Vetus Testamentum* 58(4). 469–479. <https://doi.org/10.1163/156853308X312717>.
- Hasselbach, Rebecca. 2007. Demonstratives in Semitic. *Journal of the American Oriental Society* 127(1). 1–27. <http://www.jstor.org/stable/20297208>.
- Holmstedt, Robert D. 2010. *Ruth: A handbook on the Hebrew text* (Baylor Handbook on the Hebrew Bible). Waco, TX: Baylor University Press.
- Holmstedt, Robert D. 2016. *The relative clause in Biblical Hebrew* (Linguistic Studies in Ancient West Semitic 10). Winona Lake, IN: Eisenbrauns. <https://doi.org/10.1515/9781575064208>.
- Hopper, Paul J. 1991. On some principles of grammaticization. In Elizabeth Closs Traugott & Bernd Heine (eds.), *Approaches to grammaticalization*. Vol. 1: *Focus on theoretical and methodological issues* (Typological Studies in Language 19), 17–35. Amsterdam: John Benjamins. <https://doi.org/10.1075/tsl.19.1.04hop>.
- Hopper, Paul J. & Elizabeth Closs Traugott. 2003. *Grammaticalization*. 2nd edn. (Cambridge Textbooks in Linguistics). Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9781139165525>.
- Isaksson, Bo. 2009. An outline of comparative Arabic and Hebrew textlinguistics. In Bo Isaksson (ed.), *Circumstantial qualifiers in Semitic: The case of Arabic and Hebrew* (Abhandlungen für die Kunde des Morgenlandes 70), 36–150. Wiesbaden: Harrassowitz.
- Jouion, Paul & Takamitsu Muraoka. 2006. *A grammar of Biblical Hebrew*. 2nd edn. Rome: Gregorian & Biblical Press.
- Kayne, Richard S. 2014. Why isn't *this* a complementizer? In Peter Svenonius (ed.), *Functional structure from top to toe* (The Cartography of Syntactic Structures 9), 188–231. Oxford: Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199740390.003.0007>.
- Kocher, Anna. 2022. *Complementizers on edge: On the boundaries between syntax and pragmatics in Ibero-Romance* (Open Romance Linguistics 1). Berlin: Language Science Press. <https://doi.org/10.5281/zenodo.6907848>.
- Köhler, Ludwig et al. 2001. *The Hebrew and Aramaic lexicon of the Old Testament*. Leiden: Brill.
- Kuteva, Tania et al. 2019. *World lexicon of grammaticalization*. 2nd edn. Cambridge: Cambridge University Press. <https://doi.org/10.1017/9781316479704>.
- Lehmann, Christian. 1988. Towards a typology of clause linkage. In John Haiman & Sandra A. Thompson (eds.), *Clause combining in grammar and discourse* (Typological Studies in Language 18), 181–225. Amsterdam: John Benjamins.

- Lehmann, Christian. 2015. *Thoughts on grammaticalization*. 3rd edn. (Classics in Linguistics 1). Berlin: Language Science Press. <https://doi.org/10.5281/zenodo.7520796>.
- Levinsohn, Stephen H. 2011. ׀ and ׀: Limiting and countering. *Hebrew Studies* 52. 83–105. <https://doi.org/10.1353/hbr.2011.0023>.
- Lipiński, Edward. 2001. *Semitic languages: Outline of a comparative grammar*. 2nd edn. (Orientalia Lovaniensia Analecta 80). Leuven: Peeters.
- Locatell, Christian S. 2017. *Grammatical polysemy in the Hebrew Bible: A cognitive linguistic approach to ׀*. University of Stellenbosch dissertation.
- Locatell, Christian S. 2020. Temporal conjunctions and their semantic extensions: The case of ׀ in Biblical Hebrew. *Journal of Semitic Studies* 65(1). 93–115. <https://doi.org/10.1093/jss/fgz040>.
- McCarthy, Dennis J. 1980. The uses of *w^ehinnēh* in Biblical Hebrew. *Biblica* 61(3). 330–342. <http://www.jstor.org/stable/42706879>.
- Meyer, Esias E. 2001. The particle ׀, a mere conjunction or something more? *Journal of Northwest Semitic Languages* 27(1). 39–62.
- Michaelis, Laura. 2001. Exclamative constructions. In Martin Haspelmath et al. (eds.), *Language typology and language universals*, vol. 2, 2 vols. (Handbooks of Linguistics and Communication Science 20), 1038–1058. Berlin: De Gruyter. <https://doi.org/10.1515/9783110194265-016>.
- Miller, Cynthia L. 2003. *The representation of speech in Biblical Hebrew narrative: A linguistic analysis* (Harvard Semitic Monographs 55). Boston: Brill.
- Mithun, Marianne. 2016. How fascinating! Insubordinate exclamations. In Nicholas Evans & Honoré Watanabe (eds.), *Insubordination* (Typological Studies in Language 115), 367–392. Amsterdam: John Benjamins. <https://doi.org/10.1075/tsl.115.14mit>.
- Moshavi, Adina. 2009. Two types of argumentation involving rhetorical questions in Biblical Hebrew dialogue. *Biblica* 90(1). 32–46. <http://www.jstor.org/stable/42614874>.
- Muilenburg, James. 1961. The linguistic and rhetorical usages of the particle ׀ in the Old Testament. *Hebrew Union College Annual* 32. 135–160. <http://www.jstor.org/stable/23524614>.
- Muraoka, Takamitsu. 1985. *Emphatic words and structures in Biblical Hebrew*. Jerusalem: The Magnes Press.
- Park, Grace J. 2015. ׀ from light noun to nominalizer: Toward a broader typology of clausal nominalization in Biblical Hebrew. *Hebrew Studies* 56. 23–48.
- Park, Grace J. 2016. Stand-alone nominalizations formed with *ʾāšer* and *kī* in Biblical Hebrew. *Journal of Semitic Studies* 61(1). 41–65. <https://doi.org/10.1093/jss/fgv030>.
- Park, Grace J. 2023. *Focus construction with kī 'im in Biblical Hebrew* (Linguistic Studies in Ancient West Semitic 17). University Park, PA: Eisenbrauns.
- Peeters, David, Emiel Krahmer & Alfons Maes. 2021. A conceptual framework for the study of demonstrative reference. *Psychonomic Bulletin & Review* 28(2). 409–433. <https://doi.org/10.3758/s13423-020-01822-8>.

- Redslob, M. Gustavus Mauritius. 1835. *De particulae Hebraicae כִּי origine et indole*. [On the origin and nature of the Hebrew particle כִּי]. Leipzig: Weinedel.
- Rissanen, Matti. 1991. On the history of *that*/zero as object clause links in English. In Karin Aijmer & Bengt Altenberg (eds.), *English corpus linguistics: Studies in honour of Jan Svartvik*, 272–289. New York, NY: Routledge.
- Roberts, Ian & Anna Roussou. 2003. *Syntactic change: A minimalist approach to grammaticalization* (Cambridge Studies in Linguistics 100). Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9780511486326>.
- Ross, John Robert. 1970. On declarative sentences. In Roderick A. Jacobs & Peter S. Rosenbaum (eds.), *Readings in English transformational grammar*, 222–277. Waltham, MA: Ginn and Company.
- Schoors, Antoon. 1981. The particle כִּי. *Oudtestamentische Studiën* 21. 240–276. https://doi.org/10.1163/9789004494039_017.
- Speas, Peggy & Carol Tenny. 2003. Configurational properties of point of view roles. In Anna Maria Di Sciullo (ed.), *Syntax and semantics*. Vol. 1: *Asymmetry in grammar* (Linguistics Today 57), 315–344. Amsterdam: John Benjamins. <https://doi.org/10.1075/la.57.15spe>.
- Stalnaker, Robert C. 1978. Assertion. In Peter Cole (ed.), *Pragmatics* (Syntax and Semantics 9), 315–332. New York: Academic Press. https://doi.org/10.1163/9789004368873_013.
- Staps, Camil. 2023. *Uses of Biblical Hebrew כִּי kī in Genesis, Judges, Samuel, and Ruth*. [data set]. <https://doi.org/10.5281/zenodo.8314818>.
- Staps, Camil. 2024. *The persistence of space: Formalizing the polysemy of spatial relations in functional elements*. Amsterdam: Leiden University dissertation. <https://doi.org/10.48273/LOT0673>.
- Staps, Camil. to appear. Review of: Park, Grace J. Focus construction with *kī 'im* in Biblical Hebrew. *Bibliotheca Orientalis*.
- Staps, Camil & Johan Rooryck. 2023. The interpretation of [+distal] in demonstratives and complementizers. *Linguistics* 61. 1195–1231. <https://doi.org/10.1515/ling-2022-0178>.
- Storms, G. 1966. *That*-clauses in Modern English. *English Studies* 47. 249–270. <https://doi.org/10.1080/00138386608597258>.
- Thorion, Yochanan. 1984. *Studien zur klassischen hebräischen Syntax*. [Studies on Classical Hebrew syntax] (Marburger Studien zur Afrika- und Asienkunde: Serie B, Asien 6). Berlin: Dietrich Reimer.
- Van der Merwe, Christo H. J. 1993. Old Hebrew particles and the interpretation of Old Testament texts. *Journal for the Study of the Old Testament* 18(60). 27–44. <https://doi.org/10.1177/030908929301806002>.
- Van der Merwe, Christo H. J., Jacobus A. Naudé & Jan H. Kroeze. 2017. *A Biblical Hebrew reference grammar*. 2nd edn. London: T&T Clark.
- Vriezen, Theodorus C. 1958. Einige Notizen zur Übersetzung des Bindewortes *Kī*. [Some notes on the translation of the connective *kī*]. In Johannes Hempel & Leonhard Rost (eds.), *Von Ugarit nach Qumran*. [From Ugarit to Qumran] (Beihefte zur Zeitschrift für

- die alttestamentliche Wissenschaft 77), 266–273. Berlin: De Gruyter. <https://doi.org/10.1515/9783112314982-022>.
- Watts, J. Wash. 1964. *A survey of syntax in the Hebrew Old Testament*. Grand Rapids, MI: William B. Eerdmans.
- Yap, Foong Ha, Karen Grunow-Hårsta & Janick Wrona. 2011. Introduction: Nominalization strategies in Asian languages. In Foong Ha Yap, Karen Grunow-Hårsta & Janick Wrona (eds.), *Nominalization in Asian languages* (Typological Studies in Language 96), 1–57. Amsterdam: John Benjamins. <https://doi.org/10.1075/tsl.96.01yap>.
- Yap, Foong Ha & Jiao Wang. 2011. From light noun to nominalizer and more: The grammaticalization of *zhe* and *suo* in Old and Middle Chinese. In Foong Ha Yap, Karen Grunow-Hårsta & Janick Wrona (eds.), *Nominalization in Asian languages* (Typological Studies in Language 96), 61–107. Amsterdam: John Benjamins. <https://doi.org/10.1075/tsl.96.02yap>.
- Young, Ian & Robert Rezetko. 2008. *Linguistic dating of biblical texts*. London: Equinox.
- Zanuttini, Raffaella & Paul Portner. 2003. Exclamative clauses: At the syntax-semantics interface. *Language* 79(1). 39–81. <https://doi.org/10.1353/lan.2003.0105>.